RIZAL TECHNOLOGICAL UNIVERSITY
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AN AUTOMATED AQUACULTURE MONITORING SYSTEM (AUTOAQUAMONS): INTEGRATING AI FOR FISH VITALITY DETECTION AND REAL-TIME WATER QUALITY MONITORING IN AQUACULTURE AQUARIUMS

A System Developmental Study presented to the
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Bachelor of Science in Computer Engineering

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APPROVAL SHEET

This System Design project entitled “An Automated Aquaculture Monitoring System (Autoaquamons)”, prepared and presented by Lexus John H. Belisario, Ernesto B. Batang Jr., Nica Beatrice G. Corpuz, Jeric L. Durana, and John Daniel G. Furaque, in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Engineering, has been examined and recommended for acceptance and approval for Oral Examination.

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ABSTRACT

TITLE	:	An Automated Aquaculture Monitoring System (AutoAquaMons): Integrating AI for Fish Vitality Detection and Real-Time Water Quality Monitoring in Aquaculture Aquariums
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This research study, the Automated Aquaculture Monitoring System, introduces an Automatic Monitoring System for catfish that utilized the capabilities of the Internet of Things (IoT). The primary goal of the study is to reduce the fish kill on the pond relating to sudden environmental changes and improve the mortality rate of the catfish and its overall condition using automated monitoring systems that integrate different sensors, devices and parameters for accurate and real time data gathering. The system used a minicomputer, a camera, together with the 4 main sensors; pH sensor, water temperature sensor (DS18B20), dissolved-oxygen sensor, and water turbidity sensor. The pH level

sensor continuously measures and track the levels of acidity and alkalinity of the water, the temperature sensor captures real-time temperature data of the aquarium/pond, the dissolved-oxygen sensor measures the level of the oxygen that is present in the water, and the turbidity sensor estimates the turbidity of the water, this device helps to identify the clarity and particle content of the water. The output readings are relayed to the database and will be displayed instantaneously on the website and can be monitored remotely by the user. This suggests that the use of the system aids the catfish farmers and hobbyists in saving time and energy, and most importantly it helps them to predict and monitor the pattern of environment of their pond/aquarium.

Keywords: *pH Level, Internet of Things, Water Temperature Sensor, Dissolved Oxygen Sensor, African Catfish, Water Turbidity Sensor*

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CHAPTER I

1.1 INTRODUCTION

Fishkeeping has become increasingly popular, with both commercial and ornamental aquaculture thriving as sources of decoration, hobby, and income. However, manually monitoring aquarium or pond conditions can be challenging (Muyot et al., 2019). This study focuses on developing an automated system for African Catfish aquaculture, enabling remote monitoring and management. In the Philippines, aquaculture plays a crucial role in food security, providing sustenance, income, and livelihoods to millions. Yet, challenges such as insufficient harvests due to heat and drought persist (National Academy of Science and Technology (NAST), 2017). To address these issues, the proponents employ advanced sensors to monitor key environmental parameters such as: temperature, pH, turbidity, and dissolved oxygen.

These sensors streamline tasks and enable wireless tracking of catfish production from remote locations. By integrating these technologies, the proponents aim to enhance the efficiency and sustainability of African Catfish aquaculture, ensuring optimal conditions for growth and productivity (Akosuga et al., 2024). This innovative approach not only supports the local fishing industry but also contributes to the broader goal of sustainable aquaculture practices. Through this study, the proponents seek to demonstrate the potential of technology in transforming traditional fish farming into a modern, efficient, and profitable venture. (Amora et al., 2020).

1.2 BACKGROUND OF THE STUDY

In today's world, the concept of an automated aquarium is gaining traction, offering a modern solution to the challenges of traditional fishkeeping. Monitoring key parameters such as pH level, temperature, turbidity, and dissolved oxygen levels are essential for maintaining a healthy aquatic environment. However, manual checks can be time-consuming and require significant dedication (Afifah, 2019). This study proposes an Internet of Things (IoT)-connected system designed to monitor the upkeep of aquariums, specifically focusing on African Catfish. By utilizing interconnected devices, the system provides real-time data through the website of the system, allowing for efficient oversight of the aquarium's conditions.

The system continuously tracks changes in the water's physical state, ensuring optimal conditions for the catfish. By analyzing collected data, it can identify potential causes of catfish mortality, offering insights into environmental factors that may impact fish health. This approach not only reduces the manual effort required but also enhances the understanding of aquaculture dynamics, contributing to more sustainable and effective fish farming practices. Through this innovative system, the proponents aim to transform traditional aquaculture into a more efficient and insightful process, supporting catfish owners, business owners, hobbyists and commercial catfish farmers in maintaining thriving aquatic environments (Apribowo, 2019).

1.3 STATEMENT OF THE PROBLEM

1.3.1 General Problem

Many ponds experience periodic fish kills, primarily due to the absence of an effective monitoring system to oversee the health and well-being of the fish. This study aims to reduce the fish kill rate and enhance the overall condition of catfish by implementing a comprehensive monitoring solution.

1.3.2 Specific Problem

1. How can the system accurately monitor, and display data gathered from the aquarium or pond?

2. How can the system provide long-term benefits to the aquarium or pond by notifying users of critical changes?

3. How can the system assist users in enhancing the overall condition of African Catfish by providing timely notifications and data analysis?

1.4 OBJECTIVE OF THE STUDY

The proposed system allows users to remotely monitor the conditions of the catfish and water. It provides notifications of any significant changes to all users.

1.4.1 Specific Objectives

The specific objectives of the proposed system are as follows:

1. Develop a system to monitor the condition of African Catfish in the aquarium.
2. Identify possible causes of fish mortality based on water and environmental conditions.
3. Record and report the number of alive and dead catfish.

4. Enable catfish farmers to remotely monitor the condition of catfish and water in their ponds.
5. Provide valuable data to help improve water conditions.

1.5 SYSTEM DESIGN PARADIGM

As shown in Figure 1.1, is the Input-Process-Output Model, also known as the IPO Model, that will represent how the designed system the proponents have proposed, will work, and the flow of process in the system will be.

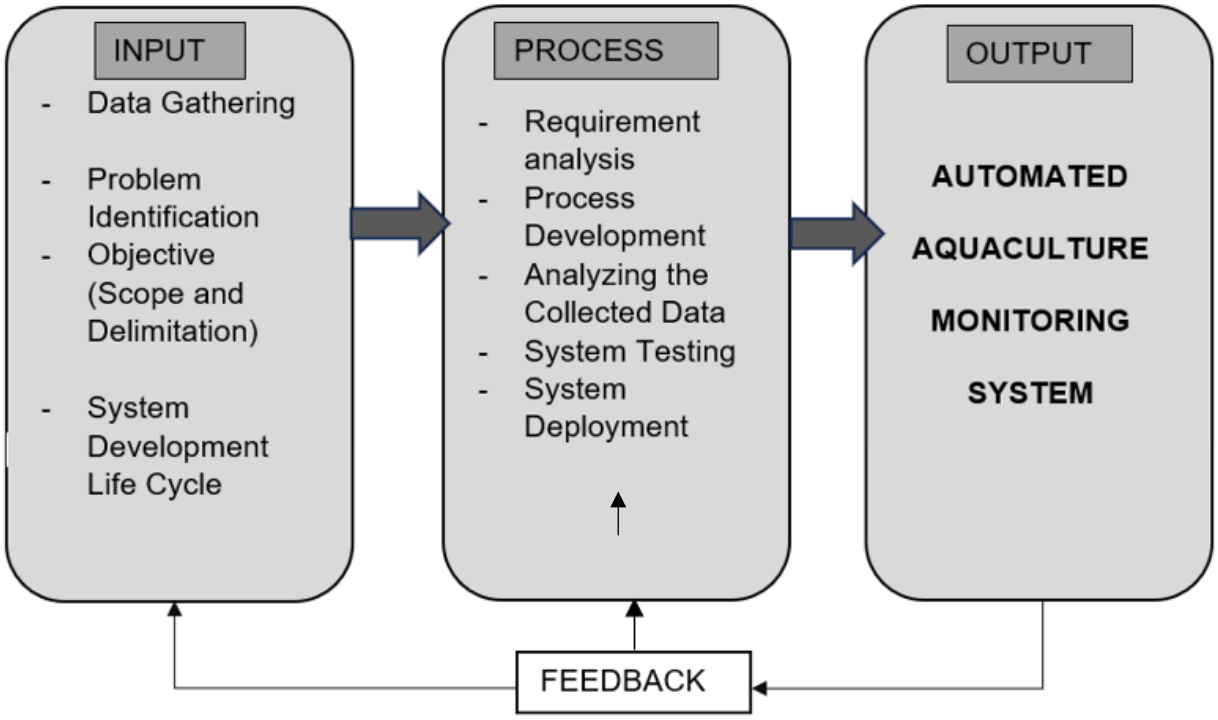


Figure 1.1: The System Design Paradigm

The following inputs listed in the figure will help the proponents who proposed the system to identify the different type of information such as Data Gathering, problem

identification, System Development Life Cycle(SDLC), and as the development of the system moves on, the proponent concluded that requirement analysis, process development, analyzing the collected data from the sensors, and system deployment are the processes to achieve the intended result as the system development progresses with SDLC's assistance in attaining the Automated Aquaculture Monitoring System.

Input - The inputs represent the flow of data and materials into the process from the outside.

Data Gathering – The proponents conducted research and studied other related studies to find the data that is vital for developing the proposed system, Fish Mortality Monitoring System

Problem Identification – At this phase, the proponent conducts a brainstorming to evaluate and determine the inadequate and true problem of the current system.

Objective – Through the acquired information, the proponents create a goal on how to handle plausible suggested replies offered by the client and real statements.

Scope – Based on the proposed system, the proponent determined the system coverage throughout the construction of this study's objective identification. At this stage, the proponent established a matrix to determine if the suggested remedy will directly address the stated problem.

Delimitation – Because all existing systems have limits that must be addressed, the proponent specified the system restriction to ensure that the intended aim is satisfied.

System Development Life Cycle – Given that this is developmental research and that the waterfall method is very relevant and dependable to achieve the project scope and delimitation, it was determined to be acceptable for creating the suggested system.

Process – The processing step includes all tasks required to effect a transformation of the inputs.

Requirement Analysis – To determine the needs or conditions to meet the requirements for the proposed idea, the proponent's system will use all necessary data.

Process Development – The proponents will conduct a last system test to ensure all the units developed in the implementation phase are integrated into a system after testing of each unit.

Analyzing the Collected Data – The proponents will analyze the conducted data that is gathered from other related studies and literatures.

System Testing – Once every unit tested during the implementation phase has been integrated into the system, a final system test will be performed.

System Deployment – The proponent has completed the system-wide testing, and the system is prepared for implementation with the intended client.

Output – The output are the data and materials flowing out of the transformation process.

Automated Aquaculture Monitoring System – The proposed system is designed and aims to develop an advanced AI controlled system that will be able to help catfish farmers monitor their catfish’s condition as well as count how many catfishes have perish, gathering the possible causes of the death of their catfishes, detecting if a catfish in the

aquarium is real or not, as well as helping the users monitor and be notified with the catfish's status and well-being remotely.

1.6 SCOPE AND DELIMITATION

The proposed system, titled "Automated Aquaculture Monitoring System (AutoAquaMonS): Integrating AI for Fish Vitality Detection and Real-time Water Quality Monitoring in Aquaculture Aquariums," aims to simplify and enhance the monitoring of African Catfish and their environment. It provides users with data on possible causes of fish mortality and the condition of the water. However, the system does not determine the exact cause of death but suggests possible reasons based on collected data. It can count the number of fish deaths and notify users remotely.

Listed here are the capabilities of the system:

1. The System is able to identify the possible causes a catfish has died.
    - High or low pH level
    - Temperature Changes
    - High or Low Oxygen Level
    - High or Low Turbidity Level
2. The system is able to analyze and detect whether it's a living or dead catfish.
3. Allows users to monitor and analyze data remotely through the website, with notifications for any catfish deaths.
4. The system is able to count how many total catfishes are inside the aquarium.

However, the proposed system also has its delimitations such as:

1. The system can only assume the possible causes of how the catfish has died based on the findings and conclusion of the system.
2. The system is only limited to a certain fish such as the African catfish only.
3. The system is only limited to detect 1 to 5 catfishes.
4. The system does not include the automated treatment for water conditions.

1.7 SIGNIFICANCE OF THE STUDY

The significance of the research the proponents proposed is significant in the context of monitoring catfish, namely monitoring the mortality rate of catfishes in a pond/aquarium. As well as finding out what the possible causes of the catfish's health decline, and the possible causes of how they perish. The people who will benefit upon conducting the study of the proposed system are:

To the catfish farmers – They will be the one to monitor the catfishes, either remotely or on-site. The proposed system will make it easier for them to analyze and monitor all the catfish's condition and the possible causes of the catfishes if one or many perish in their pond as it will be notified from them anywhere.

For future researchers – The proposed system research will help future researchers, with the concepts behind the system, when conducting similar research of their own. They can apply and develop a more innovative system and add more features to the proposed system.

For the Proponents – The proponents of this research can benefit by gaining a deeper understanding more about designing and developing an automated system, and have a

deeper understanding in aquaculture environment and practices. This study will also help the proponents to develop and enhance their expertise and skills in both programming, automation and in aquaculture.

1.8 DEFINITION OF TERMS

Aquaculture- refers to the controlled cultivation and farming of aquatic organisms, including fish, mollusks, crustaceans, and aquatic plants, under controlled conditions. It involves the breeding, rearing, and harvesting of aquatic species in various aquatic environments such as ponds, tanks, and ocean enclosures.

Fish- The fishes will be the one to be monitored in this system. They can either be pet fishes or fishes from aquaculture.

Fishkeeping - also known as aquaria, is the practice of keeping fish species in enclosed environments such as aquariums or ponds. It involves maintaining fish health, providing suitable habitats, and ensuring appropriate water quality. Fishkeeping is pursued both as a hobby and for educational or research purposes.

Fishpond- A fishpond is an artificially created or modified pond, pool, or enclosure designed for the purpose of raising and breeding fish.

Mortality - the quality or state of being a fish that is alive and therefore certain to die.

Oxygen Level - Refers to the concentration of dissolved oxygen in the water, which is crucial for the survival of aquatic organisms. Adequate oxygen levels are necessary to support respiration and metabolic processes in fish and other aquatic life.

pH Level - A measure of the acidity or alkalinity of the water, expressed on a scale from 0 to 14. A pH level of 7 is neutral, below 7 is acidic, and above 7 is alkaline. Maintaining an appropriate pH level is essential for the health and well-being of fish.

Proponents- are the people who proposed the idea and the research itself that will be used in developing the designed system.

Temperature Level – Refers to the degree of heat present in the water, which affects the metabolic rate, growth, and reproduction of fish. Different species of fish have specific temperature ranges that are optimal for their survival and growth.

Turbidity Level - A measure of the cloudiness or haziness of water caused by suspended particles. High turbidity levels can reduce light penetration, affecting photosynthesis in aquatic plants and potentially harming fish by clogging their gills.

CHAPTER II

2.0 REVIEW OF RELATED LITERATURE AND STUDIES

In this chapter, the review of related literature and studies, which is a cornerstone in research, discovers knowledge gaps, develops research questions, and synthesizes data from several sources to determine the study's context and importance. This study also identifies theoretical frameworks and procedures from past research, which helps to create a strong research design.

2.1 FOREIGN LITERATURE AND STUDIES

This chapter introduces the concepts, generalizations or conclusions, techniques, and other concepts. This chapter's contents serve to familiarize the reader with knowledge pertinent to and similar to the subject

2.1.1 Developing an AI-based, Low-cost Surveillance System for Experimental Fish Keeping

Observation of fish behavior is a cornerstone of health assessment in research facilities. However, the presence of humans can interfere with the fish's normal behavior in different ways. In order to reduce opportunities for direct interactions of animals with humans in the laboratory setting, and minimize such interferences, a remote non-invasive monitoring system was developed from low-cost components. The system was tested with zebrafish (Danyo rerio) in 50 L glass aquariums and one externally positioned webcam with a visual field covering the entire aquarium volume. The series of images was recorded with Raspberry Pi controlled webcams, and visual data evaluation was

carried out via neural networks from the field of AI, based on the Mask R-CNN framework. Positive fish detection of 95% was achieved, confirmed by human beings on a subset of over 3000 photographs. Furthermore, software was developed to automatically load images and display AI predictions. If the predictions are incorrect, the neural network detection can be further improved by means of a human correction.

According to the researchers (Von et al., 2021), they are developing a low-cost remote non-invasive monitoring device designed to decrease direct animal-human contact in the lab. Their method was tested using Danyo rerio zebrafish in 50 L glass aquariums, using a camera with a wide vision field. Images from RaspberryPi-controlled webcams were analyzed using AI-based neural networks, specifically the Mask R-CNN framework. Fish detection was 95% accurate, confirmed by humans on over 3000 photos. Additionally, they created a software that will load photographs and present AI predictions. The amazing part is that, any incorrect predictions of the systems occur, it can be corrected by humans to enhance neural network detection.

2.1.2 Water monitoring IOT system for fish farming ponds

Fish like many living organisms have a specific tolerant range of various environmental parameters, thus fish farming of specific types of fish species requires certain conditions that have to be reached. Moreover, the people that work in the fish farming ponds have to be engaged in all day activities to maintain the living fish habitat. Therefore, monitoring and taking actions to maintain the habitat's sustainable environment for certain fish species inside of fishing ponds over distributed machine to

machine communication, which will shorten the time needed for some basic actions, is the main motivation for this paper. In this paper we present an upgrade on a functional Internet of Things (IoT) system for monitoring fish farming ponds. The IoT system consists of various sensors that measure important factors of the water quality like temperature, light intensity or water level, as well as small board computer that processes the data and sends sound and visual notifications to the fish farming manager. The current system lacks the ability to process the data to the end-user via web or mobile platform. Due to remote distance of the fish farming ponds and their location dependence of clean fresh water, one solution of this problem is using expansion module like Wivity modem to enable the end users in real time to monitor and control certain aspects of the fish farming pond IoT system. Wivity modem allows user to communicate to the IoT system via WiFi connection, cellular, LoRaWAN or satellite communication; all in one product. Later on, this module can be integrated with IoT platforms including Japser, Microsoft Azure or Amazon Web Services. For future work, we plan to expand not only the applicable services on different platforms, but also add more control modules and sensors to the existing IoT system for specific fish species.

According to the researcher (Nocheski S & Naumoski Aet al., 2018) The significance of preserving particular environmental parameters in fish farming ponds for the welfare of fish species is emphasized in the paper. The authors suggest an improved Internet of Things (IoT) system that uses a variety of sensors to evaluate water quality indicators in order to satisfy the demand for ongoing monitoring and prompt reactions.

Nevertheless, end users cannot directly access data through web or mobile platforms under the existing system. In order to enable real-time monitoring and control via WiFi, LTE, LoRaWAN, or satellite connections, the solution entails incorporating a Wivity modem. Future plans include increasing control modules and sensors to accommodate the unique requirements of many fish species, as well as connecting this module with popular IoT platforms like Microsoft Azure, Amazon Web Services, or Japser. This suggested update not only increases accessibility but also establishes the framework for future extension and customization of IoT systems for fish farming.

2.1.3 Quality Maintenance of Fish Farm: Development of Real-time Water Quality Monitoring System

Fish farming is a challenging job for producing fish without any hazard. By maintaining the water quality of the pond can safely fish production. This paper will discuss the development of a real-time water quality system to monitor fish farms. The water quality monitoring system consists of a dissolved oxygen sensor, pH sensor and water temperature sensor using CC3200 Launchpad as the microcontroller, integrated with Internet of Things (IOT) platform is set up to enable the real-time remote water monitoring system. The system is designed to efficiently monitor the water quality in fresh-water fish farm, specifically the water parameters suitable for the catfish habitat. Instead of testing the system in the real catfish farm, it is tested using a water that is simulated to the water quality for the catfish habitat.

According to the researchers (Billah et al., 2019), Fish farming poses inherent challenges in ensuring safe and hazard-free fish production, with water quality maintenance being paramount. This paper addresses this concern by presenting the development of a real-time water quality monitoring system for fish farms. The system incorporates a dissolved oxygen sensor, pH sensor, and water temperature sensor, with the CC3200 Launchpad serving as the microcontroller. Integrated with an Internet of Things (IoT) platform, this setup allows for remote real-time monitoring of water conditions. The focus is on monitoring water parameters suitable for catfish habitats in freshwater fish farms. Although their system's testing is conducted using simulated water conditions resembling those in a catfish habitat rather than in an actual catfish farm, its potential to enhance real-time monitoring in aquaculture settings is evident.

2.1.4 Design and Development of Wireless Sensor Network System to Monitor Parameters Influencing Freshwater Fishes

In this paper we have designed, developed and proposed a prototype Wireless Sensor Network (WSN) System to monitor the Fish Farm. Salinity of the fresh water is a prominent parameter and is responsible for the difference in environment from marine conditions. Salt is an effective ingredient for fishes in fighting disease, parasites and chemical poisoning. The pH is equally a key parameter which influences the environment where fishes live. Generally, the fishes can live in pH ranges from 6.0 to 9.0, but their quality of life is best between pH 7.0 to 8.0. The change in the value of pH even by a small amount will be more stressful for fish. The temperature is other parameter which

decides the value of pH and in turn affects the fishes. The developed prototype WSN system monitors these three parameters affecting the freshwater fishes. The open-source technology-ARDUINO (open electronics prototype and open source) is used to develop the system.

According to the researchers (Hampannavar et al., 2018), in order to ensure safe and efficient fish production, the challenges and problems of fish farming need a thorough approach. They stated that recognizing the critical importance of pond water quality, this research investigates the development of a real-time water quality system customized for fish farms. The CC3200 Launchpad serves as the microcontroller for the system, which includes a dissolved oxygen sensor, pH sensor, and water temperature sensor. This arrangement, when combined with an Internet of Things (IoT) platform, allows for remote real-time monitoring of water conditions. The technology the researchers of this study are developing is specifically built for freshwater fish farms and focuses on aspects that promote catfish habitat, just like our designed system. While the technology is initially evaluated using simulated water conditions similar to those found in a catfish habitat rather than in a catfish farm, the emphasis on real-time monitoring offers promise for improving the precision and efficiency of fish farming techniques.

2.1.5 Monitoring of water quality in the catfish (Clarias sp.) farming in Tuban Regency

Fish need a decent environment for their lives and livelihoods. Types of fish that have different respiratory organs, such as catfish, can survive in extreme water

conditions. However, in order to grow and develop naturally in a cultured environment, they need optimum water quality. This study aimed to determine the quality of water in catfish aquaculture ponds in three partner locations for community service activities in Tuban Regency, East Java. The method used in this research is descriptive, with 3 points of location for water sampling, namely, in Jenggolo Village (A), Tegalagung Village (B), and Campurejo Village (C), Tuban Regency.

According to the researcher (Caesar et al., 2021). The results of the study showed that the highest temperature in the pond in Campurejo Village was 30.2 ºC. The highest pH in the pond in Tegalagung Village is 7.59. The highest dissolved oxygen in the pond in Jenggolo Village was 13.4 mg/L. The highest nitrate level was in the pond in Tegalagung Village, namely, 25 mg/L. The pond in Tegalagung Village has the highest phosphate content. Furthermore, the highest Total Ammonia Nitrogen (TAN) levels are in the pond in Tegalagung Village, which is 16.6 mg/L. Based on the results, the water quality parameters were classified as great. However, some parameters are less than optimal at different locations. Water quality management needs to be developed to optimize the growth of catfish.

2.1.6 Design of Web-Based Information System on Monitoring the Water Quality of Catfish Pond

Water environmental factors that greatly affect the survival and growth of catfish include temperature and water pH with pH range of 6.5-8 and temperature of 25^0C-30^0C. The monitoring and measurement are still done manually by catfish farmers to get the

temperature and pH parameter values without knowing the exact value of these measurements. According to those problems, this final project provides an information system on monitoring the water quality of catfish pond to facilitate monitoring, namely by using a device that is remotely controlled in real-time, and contains sensors needed for measurement so that it can measure the parameters from a different place. Each node in each pond contains a measurement system using a DS18B20 temperature sensor and a SEN0161 pH sensor that can measure temperature and pH. The system is using a method with several stages, namely the stages of design, implementation and testing. The measurement results from the temperature sensor and pH sensor are displayed on the web page in the form of graphs and tables.

Based on the study of (Dwi et al., 2017) the survival and growth of catfish are significantly influenced by water environmental factors such as temperature and pH. Currently, these parameters are manually monitored by catfish farmers. This project aims to provide a remote-controlled information system for monitoring catfish pond water quality, utilizing sensors for temperature and pH measurements. Each node in the pond has a DS18B20 temperature sensor and a SEN0161 pH sensor, which are then tested and displayed on a web page in the form of graphs and tables.

2.1.7 Design of Intelligent Monitoring System for Aquaculture Water Dissolved Oxygen

One of the important factors that cause a large area death to aquaculture animals is called Rapid Hypoxia. For the safety of the aquaculture production, monitoring the

dissolved oxygen (DO) is important, an intelligent monitoring for DO of the aquaculture is designed for maintaining the DO level of the aquaculture water in a good range. Optical and polarographic DO sensor, NB-IoT low-power and narrow-band communication technology and the programmable logic controller (PLC) technology are used to build this system which realizes the full-automatic and intelligent monitoring for DO of the aquaculture water. Analyzing results from the data records of the system and the artificial sampling shows that the response time of the system reaches 5.92s.

According to the researchers (Ma. Y. & Ding W., et al 2018) without human intervention the system can control the circuit breaker of the oxygen increasing machine intelligently and can also send alarm message and feedback working status of the oxygen increasing device to the aquaculture manager and technician, replacing the traditional artificial DO monitoring and field control mode while reducing daily workload and power consumption. The system can realize intelligent 24-hour security monitoring for the DO level of the aquaculture water.

2.1.8 Water Quality Monitoring and Management for Catfish Ponds

In our Automated Aquaculture Monitoring System project, the importance of water quality in fish farming cannot be overstated. Fish, unlike land animals, live in an environment where they feed, excrete, and breathe, making water quality a critical factor in their overall health and growth. Many fish farmers, despite stocking their ponds with the right quantity of fish and providing adequate feed, still face significant losses due to a lack of understanding of water chemistry. Key water quality parameters such as dissolved

oxygen (DO), temperature, and pH must be consistently monitored and maintained within specific ranges to ensure the well-being of the fish. For example, DO levels should ideally be between 4 mg/L and saturation; levels below this can lead to reduced feed intake, slower growth, and even fish mortality.

Similarly, the temperature should be kept between 26°C and 32°C, as deviations can stress the fish, making them susceptible to diseases and affecting their growth rates. The pH level is equally important, with an optimal range of 6.5 to 7.5; values outside this range can cause stress and hinder growth. Our project aims to automate the monitoring and adjustment of these parameters, ensuring that fish farmers can maintain optimal conditions in their ponds. By doing so, we hope to reduce the risk of fish loss due to poor water quality and improve the efficiency and profitability of aquaculture operations (Towers, N.D. 2014).

2.1.9 Effect of pH on the growth performance and survival rate of Clarias gariepinus fry

African catfish (Clarias gariepinus) fry was subjected to various pH treatments to investigate their growth and survival rate. African catfish were exposed to a range of pH conditions over a period of 14 days where the growth and survival metrics are carefully monitored. In a near-neutral pH range, between 6.5 and 8, Clarias gariepinus exhibits optimal growth and survival. According to (Christopher & Anyanwu, 2016) we can determine that achieving high survival rates, this suggests that neutral pH environments support physiological developments and vitality of the african catfish due to reduced

stress on their metabolic processes. In contrast, extreme pH levels acidic and alkaline proved to be detrimental. At highly acidic levels pH 2, 3, and 10-12, the mortality rate was 100% observed within 24 hours of exposure, this indicates that the conditions are extremely lethal to the african catfish. At ph 4, survivability extends longer but total mortality occurred after 192 hours or 8 days. At ph 5, survivability reduced to 51.67%. Beyond pH 9 the survival and growth of african catfish sharply declined, these thresholds mark the boundaries where the african catfish cannot sustain life and normal development. The findings of water pH level ranging from 6.5 - 8 are successful for rearing Clarias gariepinus, while conditions below pH 5 or above pH 9, severely affect both survival and growth.

2.2 LOCAL LITERATURE AND STUDIES

The literature and studies cited in this chapter tackle the different concept, understanding, and ideas, generalization or conclusions and different development related to study of the enrollment from the past up to the present and which serves as the researchers guide in developing the project. Those that were also included in this chapter helps in familiarizing information that are relevant and similar to the present study.

2.2.1 Predicting fish kills and toxic blooms in an intensive mariculture site in the Philippines using a machine learning model

Harmful algal blooms (HABs) that produce toxins and those that lead to fish kills are global problems that appear to be increasing in frequency and expanding in area. One way to help mitigate their impacts on people's health and livelihoods is to develop

early-warning systems. Models to predict and manage HABs typically make use of complex multi-model structures incorporating satellite imagery and frequent monitoring data with different levels of detail into hydrodynamic models. These relatively more sophisticated methods are not necessarily applicable in countries like the Philippines. Empirical statistical models can be simpler alternatives that have also been successful for HAB forecasting of toxic blooms. Here, we present the use of the random forest, a machine learning algorithm, to develop an early-warning system for the prediction of two different types of HABs: fish kill and toxic bloom occurrences in Bolinao-Anda, Philippines, using data that can be obtained from in situ sensors. This site features intensive and extensive mariculture activities, as well as a long history of HABs. Data on temperature, salinity, dissolved oxygen, pH and chlorophyll from 2015 to 2017 were analyzed together with shellfish ban and fish kill occurrences. The random forest algorithm performed well: the fish kill and toxic bloom models were 96.1% and 97.8% accurate in predicting fish kill and shellfish ban occurrences, respectively. For both models, the most important predictive variable was a decrease in dissolved oxygen. Fish kills were more likely during higher salinity and temperature levels, whereas the toxic blooms occurred more at relatively lower salinity and higher chlorophyll conditions. This demonstrates a step towards integrating information from data that can be obtained through real-time sensors into an early-warning system for two different types of HABs. Further testing of these models through times and different areas are recommended.

According to the researchers (Yñiguez et al., 2020) the research tackles the global problem of harmful algal blooms (HABs) and their negative consequences on human health and fisheries. Recognizing the rising frequency and geographic extent of HABs, the researchers offer a realistic method for nations such as the Philippines, where sophisticated multi-model systems may be difficult to implement. The work effectively creates an early-warning system for forecasting two forms of HABs—fish fatalities and toxic blooms—in Bolinao-Anda, Philippines, using the random forest machine learning algorithm. The models are highly accurate, with 96.1% accuracy for fish deaths and 97.8% accuracy for hazardous blooms, demonstrating the power of empirical statistical models. A reduction in dissolved oxygen is the major predictive predictor, emphasizing its importance in predicting HAB occurrences. The findings of the study represent an important step toward integrating real-time sensor data into effective early-warning systems for various HAB occurrences, prompting additional validation across varied times and locales.

2.2.2 AquaStat: An Arduino-based Water Quality Monitoring Device for Fish Kill Prevention in Tilapia Aquaculture using Fuzzy Logic

In the Philippines, the Tilapia fish farming sector is vital to the economy in providing substantial employment, income and meeting local demand for protein sources of the Filipinos. However, the possible benefits that can be derived from this industry are at stake because of the sudden occurrences of fish kill events. This can be attributed to a wide variety of natural and unnatural causes such as old age, starvation, body injury,

stress, suffocation, water pollution, diseases, parasites, predation, toxic algae, severe weather, and other reasons. With the identified severe effects of fish kill events to the fish farmers, consumers and the fisheries industry, advanced measures and methods must be established to alleviate the adverse effects of this phenomenon. To solve the underlying problem of the water quality monitoring system to improve freshwater aquaculture, various studies were already conducted. However, these studies merely focused on the reading and gathering of water parameters. In this paper, fuzzy logic was used to come up with a model that can analyze and generate result regarding the overall quality of the water being used in Tilapia aquaculture. The water parameters considered in this paper were temperature, dissolved oxygen, and pH level. The results of the water parameter readings using the conventional method were compared to the data that were gathered by AquaStat to test its accuracy and showed no significant difference. Also, the overall water quality obtained using the conventional method was compared to the overall water quality generated by AquaStat and obtained an accurate result.

In this research, their variable for their aquaculture monitoring system is tilapia. The researchers of the study (Molato et al., 2022) stated that the Tilapia fish farming business in the Philippines is critical to the economy, contributing considerably to employment, revenue creation, and satisfying local protein needs. However, they also stated that the sector confronts hurdles as a result of recurrent fish death events caused by a variety of natural and manmade reasons. These occurrences can have serious ramifications for fish producers, consumers, and the fishing industry. Recognizing the

significance of water quality monitoring in improving freshwater aquaculture, this work applies fuzzy logic to construct a model measuring overall water quality in Tilapia aquaculture. The study compares findings acquired using the conventional technique with those provided using AquaStat, focusing on critical factors like temperature, dissolved oxygen, and pH level.

2.2.3 Machine Learning-Based Algorithm for Fish Kill Evaluation using Differential Changes of Input Parameters via Wireless Communication

The existing methods of monitoring and management of freshwater ponds to prevent fish kills cannot adapt fast enough to the frequently changing environment. Recent inventions monitor water quality and water level in fishponds in response to fish mortality related to the dissolved oxygen level, water salinity, pH levels, temperature, and many more. With the additional use of machine learning algorithms, these methods can be further improved for more energy-efficient water quality detectors. This project is a buoy-type device that is designed to transmit data through an RF transceiver module outside the set threshold embedded in the microcontroller. The buoy transmits data when there are great differential changes in the measured parameters to save energy. Then the receiver hub is equipped with a microprocessor that utilizes K-means Clustering Machine Learning Algorithm that can reconstruct as much as possible the received data. The buoys were deployed in a fishpond and tested their functionality. The sensors were calibrated and the analysis of the gathered results for the deployment showed a

significant amount of power reduction when compared with and without data transmission.

According to the researchers (Balmaceda et al., 2022) current methods for monitoring and maintaining freshwater ponds to minimize fish deaths are slow to respond to a constantly changing environment. Water quality and level monitoring innovations respond to fish mortality due to parameters such as dissolved oxygen, water salinity, pH levels, and temperature. By using machine learning algorithms, these approaches are improved, making water quality monitors more energy-efficient. The project presented here proposes a buoy-like device designed to broadcast data via an RF transceiver module when measured parameters differ considerably from the defined threshold, hence conserving energy. The received data is reconstructed by the receiver hub, which is equipped with a CPU and employs the K-means Clustering Machine Learning Algorithm.

2.2.4 IoT-BASED FISH POND WATER CONDITION MONITORING SYSTEM FOR BANGUS FARMING

According to the researchers (Basuel, R. B., & Reyes, R. V. et al., 2020) water pollution and climate change are considered the two major factors affecting water quality for fish farming. Fish kill happens yearly and it is one of the major problems encountered by the Bangus growers in Pangasinan, a province in the Philippines. Surface level temperature, water acidity (pH level) and dissolved oxygen are the usual physical parameters used in determining the water quality for fish farming. The monitoring of these parameters is done manually and periodically by farmers with the assistance of the

Bureau of Fisheries and Aquatic Resources (BFAR)by using the YSI Pro 1020, a handheld instrument used for the measurement of dissolved oxygen, temperature and the pH level of water. As a means of innovating the existing practices of fish farmers, the researchers developed an IoT-Based system that would integrate the functionalities of the YSI Pro 1020 and at the same time provide a periodic recording of water quality for analysis and processing to generate information which can be easily understood by fish farmers. A short message service (SMS) is also integrated in the system to alert the stakeholder should there be degradations on water parameters.

2.2.5 Survivability of Aqua Marine Products in Fish Ponds Through Water Quality Evaluation Using Machine Learning Algorithm

Water quality is considered as the most important factor in aquaculture production systems affecting fish health and performance. Water quality can be considered good if it relates to what aquaculture wants and needs to survive and develop, which means that aquacultured farmers must be able to understand the water quality requirements of their cultured products to ensure their fast growth and survival. Different species of fish have different and particular aspects of water quality needs in which they can live, grow and reproduce.[1] – [5] It is, therefore, imperative for fish producers to make sure that the physical and chemical conditions of pond water remain, as much as possible, within the optimum or acceptable range of the fish under culture all the time. According to the researchers (Loyola, L. G. & Lacatan L. et al., 2020) fish may show poor development, erratic behavior, and signs of disease or parasite infestations outside of these acceptable

ranges. Fish kill can occur in extreme cases or where poor conditions persist for extended periods of time.[1], [4], [6] – [8] Using water sampling to monitor water quality takes up time, while laboratory results do not show the current state of water in fish ponds, which is critical information needed by fish farmers. Water quality monitoring in fish ponds should be in real-time, analysis of water parameters must be done as soon as possible to ensure water quality and its acceptability for aquaculture products. The purpose of this study is to develop a system that can monitor water parameters in fish ponds analyze and evaluate these parameters to determine the suitability and survivability rate of aquaculture products based on water quality using machine learning algorithms regression tree and decision tree in accordance with the water quality requirements of cultured products. With the help of an Arduino microcontroller device that uses IoT (Internet of Things) technology. By implementing this study, it was found out that the system effectively helps the fish farmers to manage and help maintain the water quality of their fish ponds. It minimizes losses due to untimely solutions to water quality problems and promotes a healthy environment which helps increase the growth and survival of aquaculture products. It also increases the fish farmer harvest and income, which creates a positive impact on agricultural productivity in terms of fish farming.

2.2.6 Real-Time Water Quality Monitoring System with Predictor for Tilapia Pond

Water quality is the first most important limiting factor in pond fish production. It is also one of the most difficult production factors to understand, predict and manage. It is not just where the fish live; its quality directly affects growth rates, the fish's health and

survival. Most fish kills, disease outbreaks, poor growth, poor feed conversion efficiency and similar management problems are directly related to poor water quality. According to the researchers (Madrid, J. D. L. et al., 2018). The study and prediction of water quality is necessary to prevent serious problems that may occur during production period. In this study, water level, temperature, pH and DO levels are measured using an Arduino microcontroller. Users could receive information at predetermined intervals on preferred communication through SMS. Since integrating devices are comparatively not expensive; it usually consists of Arduino board, access to internet and relay frames and display system. With this integration system, farmers need not hire workers at their site, consequently drive down operating costs and improve efficiency. Prediction interface was developed using msSQL. The system was evaluated and it performs its intended function. In this study, a segmented moving average prediction model was found to be effective in predicting water quality parameters.

2.2.7 Monitoring of Coastal Aquaculture Sites in the Philippines through Automated Time Series Analysis of Sentinel-1 SAR Images

With the unprecedented expansion of aquaculture around the world, there is a critical need to monitor its progress. In Palawan, Philippines, coastal aquaculture is gaining momentum towards increasing fish production, a pressure that presents a challenge to the sustainability of these areas. In this paper, we explore the application of Earth observation methods to map coastal aquaculture development in Palawan and evaluate the extent of its change. The European Space Agency Sentinel-1 synthetic

aperture radar and Sentinel-2 multispectral instrument sensor data were applied in fully automatic mode to build maps of Palawan coastal aquaculture. The maps were validated using Google Earth high-resolution optical images and in situ observations in Malampaya Sound, and demonstrated a successful detection rate of 72%, while the false alarm rate was less than 7.5%. Objects only 5 m across, four times smaller than the spatial resolution of the Sentinel-1 sensors, were successfully detected using the developed methodology, thus exceeding the capabilities of other published methods that are limited to detecting large groups of aquaculture structures. The maps revealed aquaculture structures in high quantities in Malampaya Sound, Taytay Bay, and other locations in the coastal waters of Palawan, Philippines. A significant change of aquaculture spatial distribution was identified by comparing aquaculture maps generated with an interval of three years. This new automated methodology was validated as robust for mapping aquaculture objects in Palawan, Philippines, and can be applied to aquaculture studies in other regions worldwide.

According to the researchers (Kurekin, A. et al., 2022). Because of the unprecedented expansion of aquaculture around the world, there is a critical need to monitor its progress. In Palawan, Philippines, coastal aquaculture is gaining momentum towards increasing fish production, a pressure that presents a challenge to the sustainability of these areas. In this paper, we explore the application of Earth observation methods to map coastal aquaculture development in Palawan and evaluate the extent of its change. The European Space Agency Sentinel-1 synthetic aperture radar and

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2.2.8 Development of IoT-based Fish Tank Monitoring System

The aquaculture management system will be significantly improved with the adoption of recent technological advances. This study introduces the development of a fish tank monitoring system using the Internet of Things (IoT) modules with four subsystems—water quality monitoring, video surveillance of the fish tank, on-demand feeding machine, and Cloud data storage. During the evaluation, the calibrated sensors for water quality monitoring were accurate for the monitoring purpose. In contrast, the

simultaneous transmission of the sensors' gathered data from the IoT modules to the Cloud storage was 100% successful from 0.25 Mbps to 10 Mbps upload speed. The automated video surveillance was able to record 98.45% of the scheduled time.

According to the researchers (Palconit, M. G. B. et al., 2021) the implementation of recent technological advancements in the aquaculture management system, specifically the integration of an Internet of Things (IoT)-based fish tank monitoring system, demonstrates a significant enhancement in efficiency. The four subsystems comprising water quality monitoring, video surveillance, on-demand feeding, and Cloud data storage contribute to a comprehensive and automated approach. The study's evaluation highlights the precision of calibrated sensors for water quality monitoring and the flawless transmission of data from IoT modules to Cloud storage across varying upload speeds. Notably, the automated video surveillance system exhibits a high recording efficiency, capturing 98.45% of the scheduled time. This innovation showcases the potential for technology to revolutionize aquaculture management, offering precision, automation, and real-time data accessibility.

2.2.9 Design and Implementation of Real-Time Mobile-based Water Temperature Monitoring System

In this age of international trade and competition, the aquaculture industry needs to plan and implement a technology that will address issues concerning global food security. In modern aquaculture management, a remote water quality monitoring and computer-controlled intensive culture is the future trend in aquaculture. The objective of

this research is to design and develop a real-time mobile-based water temperature monitoring system capable of decreasing the reliance on manpower at the monitoring site to reduce the cost and to assess fish production cycle and fish grow-out system. The system implementation resulted in a monitoring system that collects the current water temperature from the core-controller in real-time. Also, the system provides and displays information that includes normal range, maximum, minimum, average and findings of the collected temperatures. The results obtained in this study have shown the ability of data acquisition in the remote and real-time detection of water temperature accurately and efficiently.

Based on the study of (Bokingkito Jr, P. B. & Orven, E. L et al., 2017) the aquaculture industry needs to implement technology to address global food security issues. This research aims to design a real-time mobile-based water temperature monitoring system that reduces manpower reliance, costs, and assesses fish production and grow-out systems. The system collects real-time water temperature data from a core-controller and displays information like normal range, maximum, minimum, average, and findings. The results show accurate and efficient remote and real-time water temperature detection, providing decision support to fishers to avoid distress and achieve the optimal water temperature range.

2.2.10 Design and Implementation of AquoSense – a Water Quality SensorSystem

Aquaculture has become an effective breeding and farming method for aquatic plants and animals, but is also rendered vulnerable by increasing weather fluctuations

and unexpected environmental changes that affect water quality. Real-time detection of fluctuations in water quality parameters is important for executing appropriate and timely operational responses to minimize losses due to fish kills or a collapse of a harvest cycle. This necessitates continuous monitoring which, at present, is mostly done manually, aperiodically, inconsistently, haphazardly, and oftentimes reactively. To address these pain points, the proponents have designed and developed AquoSense - a low-cost and easy-to-use water quality sensor system. AquoSense can be automated or pre-programmed to regularly measure parameters such as pH, dissolved oxygen (DO), electrical conductivity (EC), and temperature; and monitor the derived parameters salinity, total dissolved solids (TDS), and specific gravity (SG). Tests done for pH and EC sensor probes revealed that it can be comparable to a commercial-grade water quality measuring device. AquoSense can store data locally, and/or transmit these to a local mobile device or to the internet cloud. AquoSense comes in two versions: handheld and submersible, and both have been tested in the laboratory and deployed in aqua field conditions.

According to researchers of this study (Monje, J. C. et al., 2019) aquaculture has proven to be an effective method for breeding aquatic plants and animals, yet its vulnerability to unpredictable environmental changes, especially in water quality, poses significant challenges. Timely detection of fluctuations in water parameters is crucial to mitigate losses, such as those from fish kills or disrupted harvest cycles. The current manual and sporadic monitoring methods lack consistency and are often reactive. In

response, the proponents have introduced AquoSense—a cost-effective and user-friendly water quality sensor system. Capable of automated or pre-programmed measurements for pH, dissolved oxygen, electrical conductivity, and temperature, AquoSense monitors derived parameters like salinity, total dissolved solids, and specific gravity. Tests indicate that the pH and electrical conductivity sensor probes compare favorably with commercial-grade devices with options for data storage and transmission to local devices or the cloud, AquoSense, available in handheld and submersible versions, has undergone successful laboratory testing and real-world deployment in aquaculture settings. This innovation addresses the need for continuous, reliable monitoring in aquaculture operations.

2.1.11 Development of Dissolved Oxygen Monitoring System for Fish Ponds

Dissolved oxygen is probably the single most important water quality factor that pond managers need to understand. The purpose of the present study is to create a program that helps fish pond owners and aquaculturists in producing high quality fishes by maintaining normal levels of dissolved oxygen in fish ponds. Also, through this system, will solve the long-time problem of fish-kill in fishponds. The study was tested in ponds and rivers in Kawit and Noveleta, Cavite. The researchers joined the Bureau of Fisheries and Aquatic Resources during the actual rounds of monitoring water quality in the said area. Of the 20 respondents, all gave excellent rating for the criteria included in the survey, the speed, user friendliness, reliability, and security. The user friendliness got the highest score, followed by reliability, security, and speed which got the lowest score. Based on

the findings of this paper, the program's speed, reliability, user friendliness, and security are all ranked excellent by the respondents.

Based on the study of (Garcia, M. et al., 2013) focuses on developing a program to aid fish pond owners and aquaculturists in producing high-quality fish by regulating dissolved oxygen levels. The program aims to address the persistent issue of fish kills in ponds. Field-tested in Kawit and Noveleta, Cavite, the study involved collaboration with the Bureau of Fisheries and Aquatic Resources for real-time water quality monitoring. The survey, conducted among 20 respondents, indicates unanimous excellent ratings for the program's speed, user-friendliness, reliability, and security. Notably, user-friendliness received the highest score, followed by reliability, security, and speed. The overall findings underscore the program's excellence in terms of speed, reliability, user-friendliness, and security, showcasing its potential to enhance fish farming practices and mitigate fish-kill incidents.

2.2.12 Automated Aquaculture System That Regulates Ph, Temperature and Ammonia

Fish ponds exposed to the weather is the current method of raising tilapia in the Philippines and the methods for measuring pH level, temperature, dissolved oxygen and ammonia are limited to manually using a chemical kit. Manually regulating the water quality from the risk of harmful situations is the current system resulting from unsafe levels of temperature, Ph, and dissolved oxygen. The study aims to create a system that

automatically measures these methods and takes advantage of electronics for pH, temperature, and dissolved oxygen while computing the ammonia factor.

According to the researchers (Africa, A. D. M. et al., 2017) it allows the user to measure levels of the said parameters at any given time, process, and send the data to a LabVIEW database and use the data to automatically take corrective actions against harmful levels of pH, temperature, dissolved oxygen and ammonia while notifying the user through SMS. The proponents of this study built the prototype and tested it on two different trials of 50 fingerlings each in a 1 cubic-meter glass aquarium.

2.2.13 Automated Aerator System for Controlling Dissolved Oxygen and Monitoring Acidity and Temperature on Aquatic System.

Designing and developing a system that monitors pH level and temperature with dissolved oxygen in an artificial aquatic system is the main purpose of the study. The sensors were submerged inside the aquatic tank to monitor their values and using experimental methods in testing the functionality of the different sensors were successful as the system monitors coming from the factors. The information coming from pH level, temperature and dissolved oxygen reading were displayed to an android phone to bluetooth serial module then displayed to the system software.

Based on the study of (Cuaton, M. J. A. et al., 2018) Valve 1 is the intake valve and Valve 2 is the discharge valve, when acidity reaches 6.4 or below in readings the two valves open. Valve 2 flushes out the acidic water and valve 1 replaces it with non-acidic water. When the acidity of the water reaches 6.5 up to 9.0, the two valves close which

signifies that acidity of water is normal. A minimum of 3 psi is needed to operate the valve effectively and since most species cannot tolerate acidic water, the pH reading is triggered. The results show that by replacing the water, not only the acidity was lowered but the dissolved oxygen content also was either maintained or increased.

2.2.14 AN INNOVATIVE REAL-TIME WATER QUALITY MONITORING SYSTEM FOR AQUACULTURE APPLICATION

One of the most promising industries in the agricultural sector is aquaculture and fish is an aquaculture's contribution as a source of protein for people around the world. However, issues confronting small-scale aqua farmers is the current practice of conventional water quality monitoring which is tedious and time consuming. Laboratory-based tests that require fresh supplies of chemicals, qualified staff and water samples are the current evaluation method. An innovative real-time water quality monitoring for aquaculture application that utilized electronics sensors, microcontroller and SMS technology for notification purposes is proposed. There are four parameters, monitored namely: pH, temperature, dissolved oxygen and ammonia. The results show that there are minimal errors in the values of the parameters in the proposed device as compared to the standard device and function to its purpose with a high degree of accuracy. The proposed device is a potential innovative solution to small-scale aqua farmers in mitigating fish kill yielding to increase in production.

According to the study of Fowler et al. (1994), pH, temperature and DO must be checked continuously and on a direct basis since they tend to change quickly and have

a significant adverse effect on the system if allowed to operate out-of-range. Also, an essential element in determining water quality which impact fish behavior and health is ammonia (Pillay et al., 2005). Many monitoring systems in fish farming management have been remotely developed. The study of Nasirudin et al. (2011) proposed a smart system to monitor the water quality remotely via SMS. It monitored and recorded real-time data of pH level and DO level. In the study of Silva et al. (2011), they used WiMAX and sensors like pH, conductivity, temperature, ORP and DO.

2.2.15 Automated Water Quality Monitoring for Aquaponics Applied to Vertical Farming

Aquaponics is the process of growing plants and fish together in one system, relying on the biotic relationship between the fish and bacteria. Fish raised in aquaponics requires good water quality conditions as it is a major factor for the optimal growth of the fishes. However, the water quality parameters of the aquaponics in the Philippines are not monitored and controlled which affects the growth of the fishes. The study focused on the development of a device that automatically monitors and controls the water quality parameters of aquaponics in application of vertical farming. The design device is based on the ideal water quality parameters suitable for growing tilapia such as ammonia, dissolved oxygen, pH level, temperature and turbidity. It is capable of monitoring and measuring when one of the water quality parameters is not within its ideal value. The device test results are based on its accuracy and reliability, it shows that the pH level and

temperature sensors are 98.42% and 97.07% accurate. The device is also 100% reliable as it functions based on its expected output.

According to the researchers (Garcia, S M. P. & Santos, C. L. B. et al., 2018). The device was able to measure values of ammonia, dissolved oxygen, pH, temperature and turbidity of the water. The results of the tests proved that the sensors of the device are all accurate and reliable, proving its capability to maintain and monitor the water quality of aquaponics in a vertical farm. The accuracy and reliability can be further improved by using an ammonia sensor-compatible to a microcontroller board to directly monitor the ammonia in water. Provision of a user interface for better measurement display and for configuration of water quality parameters.

Table 2.3.1 Synthesis of Related Literature

FEATURES	A	B	C	D	E	F	G	H	I	J
Able to be used Remotely	✓	✗	✗	✓	✓	✓	✗	✓	✓	✓
Able to monitor Aquarium Temperature, Oxygen, pH Level, and turbidity.	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓
Able to detect how many fishes are on the aquarium	✗	✗	✓	✗	✗	✗	✗	✗	✓	✓
Able to identify the possible causes a fish have died	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓
Able to notify user if a fish perishes	✗	✓	✓	✗	✗	✗	✓	✗	✗	✓
AI-integration	✗	✓	✗				✓	✓	✓	✓
Uses Machine Learning	✗	✗	✓	✗	✗	✗	✗	✗	✓	✓
Uses React JS and Python(flask)	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓

LEGEND:

System A - IoT-BASED FISH POND WATER CONDITION MONITORING SYSTEM FOR BANGUS FARMING

System B - IMPLEMENTATION OF SMART AQUARIUM MONITORING SYSTEM

System C - Developing an AI-based, Low-cost Surveillance System for Experimental Fish Keeping

System D - The smart monitoring and automation control system for fish aquarium based on internet of things technology

System E - IOT BASED AUTOMATED FISH TANK MONITORING SYSTEM

System F - Quality Maintenance of Fish Farm: Development of Real-time Water Quality Monitoring System

System G - AquaStat: An Arduino-based Water Quality Monitoring Device for Fish Kill Prevention in Tilapia Aquaculture using Fuzzy Logic

System H - Machine Learning-Based Algorithm for Fish Kill Evaluation using Differential Changes of Input Parameters via Wireless Communication

System I - Development of an IoT-based Intensive Aquaculture Monitoring System with Automatic Water Correction

System J - Automated Aquaculture Management System (AutoAquaManS): Integrating AI for Fish Vitality Detection and Real-time Water Quality Monitoring in Aquaculture Aquarium

CHAPTER III

RESEARCH METHODOLOGY/DESIGN

This chapter introduces the process researchers created, including tools used throughout the data collection process to develop the study, the study site, and the developed process model. A software system is used to evaluate the planned system.

3.1 Research Instruments

This chapter introduces the process that researchers developed including tools used throughout the data collection process to develop the study and process model developed. The study aims to understand and counteract fishkeeping problems with the combination of data collected. Questionnaires were provided to fish keepers and non-fish keepers to learn more about their opinions and insights in this field.

3.2 Locale of the Study

The study was conducted at Binangonan Rizal, this place is selected for knowing the efficiency of the said study in terms of ideal fish keeping environment. The research aimed to observe, address, and minimize the problems regarding fishkeeping.

3.3 Population of the Study

To select those who are available and willing to participate in this study, a convenience sample was used by the researcher. The Study will focus on people who were previously or currently fish farmers/hobbyists who reside in the Philippines.

3.4 Survey Data Gathering

To ensure that the data gathered are valid and accurate, the researcher used the data gathering tools listed below:

Observation

Observation is the collection of Primary Data through observing the proposed system by watching the subject's behavior, and events, or noting physical characteristics in their natural setting.

Research Methodology

To compute the weighted mean for each criterion and the overall system evaluation, the **Weighted Average Formula** was applied:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i \cdot w_i}{\sum_{i=1}^{n} w_i}$$

$$\bar{x} = \frac{(x_1 \cdot w_1) + (x_2 \cdot w_2) + \ldots + (x_n \cdot w_n)}{w_1 + w_2 + \ldots + w_n}$$

where xi represents individual ratings, wi represents weights (assumed equal), and n is the number of ratings. Since all ratings carry equal weight, the formula simplifies to the arithmetic mean. For **System Accuracy and Performance**, the ratings (4.7, 4.2, 4.0, and 4.4) resulted in a weighted mean of **4.325**. The **Maintainability of the System** had ratings of 4.5, 4.3, 4.2, and 4.4, leading to a weighted mean of **4.35**. The **Reliability of the Proposed System**, based on five ratings (4.2, 4.4, 4.2, 4.3, 4.5), resulted in **4.32**.

For **Functionality**, the ratings (4.4, 4.3, 4.5, and 4.6) produced a mean of **4.45**. Meanwhile, the **Operability of the System**, rated at 4.2, 4.2, 4.1, and 4.2, resulted in **4.175**. Finally, the overall weighted mean was computed as the average of all category means, leading to a final system rating of **4.324**. This confirms that the **Automated Aquaculture Monitoring System** received high satisfaction ratings across all key performance aspects, indicating strong user approval.

Survey Questionnaire

The researchers will give out survey questionnaires to the fish farmers and hobbyists that gather statistical information about the attributes, attitudes, or actions of a population through a structured set of questions. The participants' answers were interpreted using a 5-point Likert scale based on ISO 25010. The Likert scale had a sequence of statements, each with a scale ranging from 1 to 5. A rating of 5 indicated the highest or positive response, while a rating of 1 indicated the lowest or negative response. The scale's middle point, 3, denoted a neutral response. The following is the five-level Likert scale which is used to interpret the data gathered: The ISO 25010 is an International Standard that outlines how to establish and maintain a quality management system.

Table 3.4.1 Survey Rating Scale

Rating scale	Numerical values	Descriptive value
5	4.21-5.00	Very Satisfied
4	3.41-4.20	Satisfied
3	2.61-3.40	Neutral
2	1.81-2.60	Dissatisfied
1	1.00-1.80	Very Dissatisfied

Results

The researchers divided the survey questionnaires into 3 different categories, the first set of questions is about Target Respondents (Fish Farmer/Hobbyists). The second set of questions tackles the efficiency and accuracy of the website, and the last set addresses the proficiency and reliability of the system in terms of data gathering and monitoring of the fish and its environment inside the aquarium/pond. The questions provided were answered by 17 respondents.

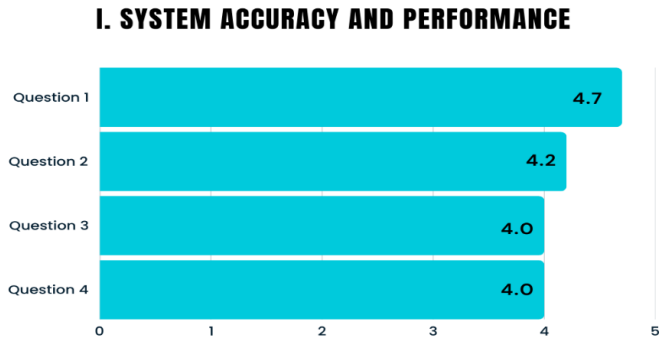


Figure 3.4.1 System Accuracy Survey Ratings

The graph indicates that selected respondents are highly satisfied with the Automated Aquaculture Management System's accuracy and performance with the

weighted mean of 4.325. This suggests that the system performs well, reads and displays the data accurately.

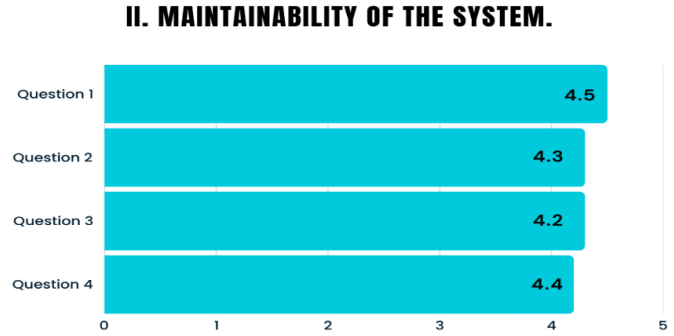


Figure 3.4.2 System Maintainability Survey Ratings

This graph illustrates that the maintainability of the system was rated 4.35, showing that users find it easy to upgrade and modify without compromising performance. The system's reusability was also acknowledged, meaning it can be integrated into other tools or adapted for future improvements with minimal issues.

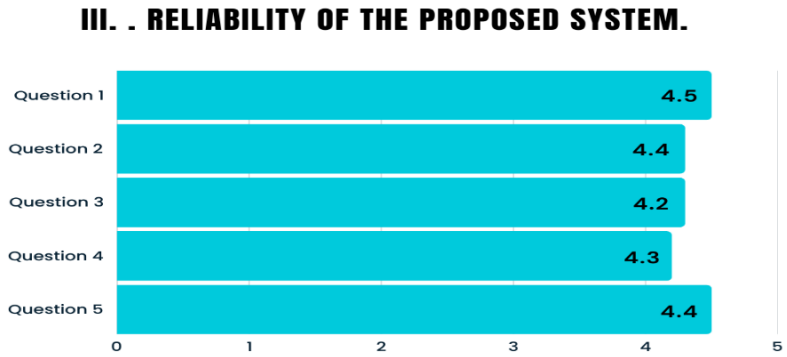


Figure 3.4.3 System Reliability Survey Ratings

This graph illustrates that the reliability of the proposed system was rated 4.32, demonstrating that users find it dependable in terms of uptime, performance consistency, and recovery from failures. This suggests that the system operates without major faults

under normal conditions, ensuring uninterrupted monitoring of the aquaculture environment.

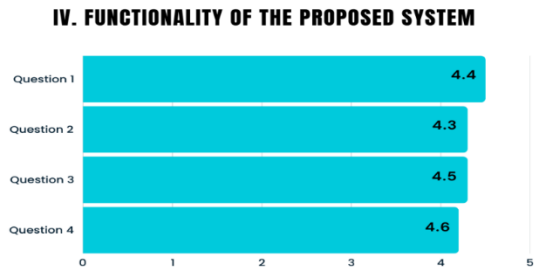


Figure 3.4.4 System Functionality Survey Ratings

This graph illustrates that the functionality of the system was rated 4.45, this indicates that the system is functioning well in terms of its overall features, such as gathering of data, monitoring of the catfish, and displaying it on the website.

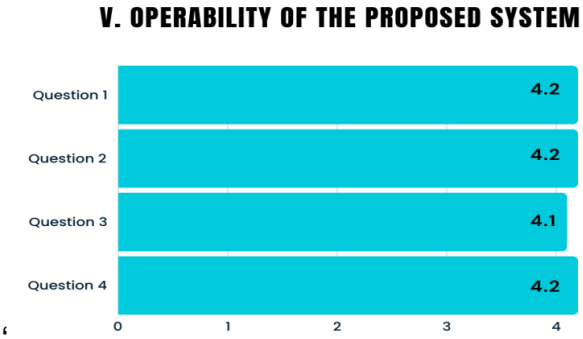


Figure 3.4.5 System Operability Survey Ratings

This last graph illustrates the operability of the system receiving a 4.175 rating, the lowest among the criteria but still indicating overall satisfaction. Users appreciate the dashboard's ease of use and remote monitoring capabilities. However, there may be room for improvement in data accuracy, particularly in the system's predictions related to catfish mortality.

Table 3.4.2 Survey Questionnaire Tallied Results

QUESTIONS		
I. SYSTEM ACCURACY AND PERFORMANCE	WM	VI
1. Rate the accuracy of the system maintaining the water pH level, temperature, turbidity and oxygen.	4.7	Highly Satisfied
2. Rate the accuracy of the system counting the number of living catfish in the aquarium.	4.2	Satisfied
3. Rate the accuracy of the system counting the number of dead catfish in the aquarium.	4.0	Satisfied
4. How satisfied are you with the system's overall accuracy?	4.4	Satisfied
Weighted Mean	4.325	
II. MAINTAINABILITY OF THE SYSTEM		
1. How satisfied are you with the system intended to change one or more parts to higher end components effectively increase its efficiency under normal operation?	4.5	Highly Satisfied
2. How satisfied are you with the re-usability of the system to be used as a tool in more than one system or in building other tools.	4.3	Highly Satisfied
3. Rate your satisfaction with the system being modified without defects or degrading the quality of Automated Aquaculture Management System.	4.2	Satisfied
4. Rate your satisfaction with the system's overall maintainability.	4.4	Highly Satisfied
Weighted Mean	4.35	
III. RELIABILITY OF THE PROPOSED SYSTEM		
1. How satisfied are you with the system's uptime and availability?	4.2	Satisfied
2. Rate your satisfaction with the consistency of the system's performance.	4.4	Satisfied
3. How satisfied are you with the system's ability to recover from failures or errors?	4.2	Satisfied
4. How satisfied are you with the system's product or component to perform specific functions without fault under normal operation?	4.3	Satisfied
5. How satisfied are you with the system's overall reliability?	4.5	Highly Satisfied
Weighted Mean	4.32	
IV. FUNCTIONALITY OF THE PROPOSED SYSTEM		
1. How satisfied are you with the system's functionality to complete all tasks and meet the user's objective?	4.4	Highly Satisfied
2. How satisfied are you with the system's accurate results with the needed degree of precision?	4.3	Satisfied

3. How satisfied are you with the system's ability to accomplish specified task and objectives?	4.5	Satisfied
4. Rate your satisfaction with the system's overall functionality.	4.6	Highly Satisfied
Weighted Mean	4.45	
V. OPERABILITY OF THE PROPOSED SYSTEM		
1. How satisfied are you with the dashboard used to control and monitor the system? - User - Interface	4.2	Satisfied
2. How satisfied are you with the ability to monitor the system remotely from anywhere using the IoT Application? - Remote Monitoring	4.2	Satisfied
3. Rate your satisfaction with the system's ability to collect data accurately, including its predictions regarding to the cause of death of the catfish - Data Accuracy	4.1	Satisfied
4. Rate your satisfaction with the system's overall operability.	4.2	Satisfied
Weighted Mean	4.175	
Overall Weighted Mean	4.324	

The overall satisfaction with the Automated Aquaculture Monitoring System is high, with an overall weighted mean of 4.324, indicating that users generally agree on its reliability and effectiveness. The system's functionality (4.45) received the highest rating, showing that it successfully meets user objectives with accurate and precise results. Maintainability (4.35) and system accuracy (4.325) were also rated positively, highlighting that users find it efficient, reusable, and capable of monitoring water parameters and detecting catfish accurately. Reliability (4.32) scored well, reflecting consistent system uptime and recovery from failures. The lowest, though still strong, rating was for operability (4.175), indicating that while users are satisfied with its dashboard and remote monitoring capabilities, there may be minor areas for improvement in data accuracy.

Overall, the results suggest that the system is well-received and performs effectively in its intended functions.

3.5 Research Design

The systems development life cycle (SDLC) is a conceptual model used in project management that describes the stages involved in an information system development project, from an initial feasibility study through maintenance of the completed application. Software development is an organized process that thrives to deliver products in faster, better and cheaper ways. Agile methods strive to deliver small sets of software features to customers as quickly as possible in short iterations. As part of this paper scope most commonly used methods will be examined from the angle of their applicability, strengths and weaknesses and their adoption in industry. This will lead us to find benefits, limitations and difficulties in agile software development.

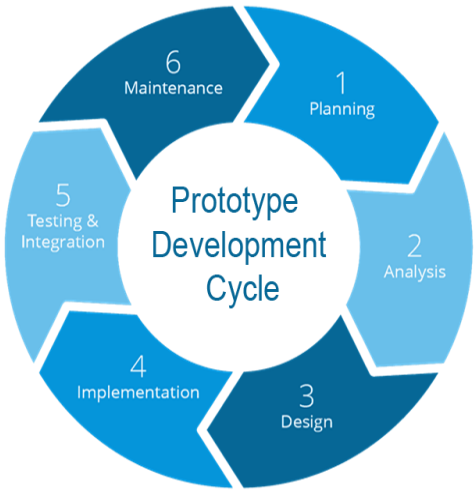


Figure 3.5: Agile software development life cycle model

This model is known as the Agile SDLC model. It combines iterative and incremental process models, focusing on process adaptability and user satisfaction by rapidly delivering working software products. An iterative approach is taken and a working software build is delivered after each iteration. Each build is incremental in terms of features; the final build holds all the features required by the customer. The Agile thought process started early in software development and started becoming popular with time due to its flexibility and adaptability.

3.6 Unit and Components Testing

Table 3.6.1 *User Web Application Testing*

USER WEB APP	READINGS	TIME	REMARKS
Able to display the current environmental status of the aquarium.	Temperature (°C), Oxygen (mg/L), pH, and Turbidity (NTU)	Every 2 seconds	The system successfully displays the sensor readings
Able to display the report of the possible reasons of a catfish's death	Data gathered from the previous 3 hours	Upon each catfish death	The system successfully notifies the user every time the catfish dies
Able to display the number of alive and dead catfishes	Count of alive and dead catfish	Upon each detection of any catfish	The system successfully displays the number of catfishes
Able to display the weekly graph of each data gathered	Data gathered from the previous week	Every 1 week	The system displays the graph of each sensor readings and automatically displays the data on the graph.
Able to manually download all of the data gathered depending on the time	Hourly, Daily, and Weekly Data gathered	1 Second	The system downloads the reported data on each date the user selects.

Table 3.6.1 presents the results of testing each feature and component of the web application for the Automated Aquaculture Management System. The testing covered various scenarios to ensure that all functions and components of the web application are

fully operational and perform as expected. Each feature was verified for accuracy and reliability, confirming that the system can effectively monitor and report on the aquarium's conditions and catfish status.

Table 3.6.2 *Website Testing*

Log In	Status	Remarks
Verifies username and password	The system verifies the username and password	Successful
System Monitoring Parameter		
View Water Temperature	The system shows the water temperature	Successful
View Water pH level	The system shows the water pH level	Successful
View Water Turbidity	The system shows the water turbidity	Successful
Water Temperature Chart		
Monitor the temperature of the water	The system monitors the temperature of water	Successful

Table 3.6.2 shows that the Automated Aquaculture Monitoring System website is fully functional and can generate a report.

Data Analysis

Content analysis will be used to analyze the data gathered categorizing verbal or behavioral data to classify, summarize, and tabulate the data. Researchers quantify and analyze the presence, meanings, and relationships of such words and concepts, then make inferences about the messages.

Operation and Testing Procedure

A consolidated requirement should meet the client's specifications. Operating and testing procedures were performed to verify the intended output met the desired outcome before implementation.

Operation Procedure

Website Administrator

1. Read and follow the manual before using the system
2. Redirect to the website.
3. The system should take necessary precautions.
4. Create and enter the password.
5. Navigate at your discretion.

Evaluation Procedure

Evaluation is an essential part of the software development process, as it helps determine how well the system meets its intended goals and user needs. This process involves a thorough examination of the software's features, performance, and usability. One of the key objectives during evaluation is to ensure that any form of bias—whether in the design, data, or interpretation—is identified and minimized to produce accurate and reliable results.

To support the evaluation, the researcher also distributed questionnaires to selected respondents. These questionnaires were carefully designed to gather insights, feedback, and observations from users who interacted with the software. The responses collected served as the foundation for assessing various aspects of the system, helping the researchers gain a more comprehensive understanding of its strengths and areas for improvement.

CHAPTER IV

This chapter introduces the system, Automatic Aquaculture Monitoring System, where it includes all the data that was gathered, System Testing and Data Collection, Non-functional System Requirements, Screen Transition, and User Website in undertaking the development of the project.

4.1 Overview

The Automated Aquaculture Monitoring System provides fish keepers, specifically catfish owners, an efficient and effective way of taking care of the catfishes, and prolonging their life, making their farming of catfish much easier and making the catfish healthier. The system is designed to provide the catfish farmers, hobbyists, pet owners, and business owners better and accurate data in terms of how their catfishes behave inside the aquarium and its mortality rate. That way, all of them can easily figure out the cause of their fishes' death, and they can easily find the way to prevent any issues inside the aquarium or pond that may cause a fish to die. This system will make catfish keepers farming methods much more efficient and effectively improve the catfish's quality.

4.2 User Website

As shown here in Figure 4.2, the dashboard of the website of the Automated Aquaculture Monitoring System, where the user can monitor the live status of the Temperature, Oxygen, pH, and Turbidity level of the water. If such fluctuations occur, the system through the website will notify the user that there is a fluctuation and the system

will recommend the user to change the water immediately to prevent the catfish from being stressed which can lead to its mortality

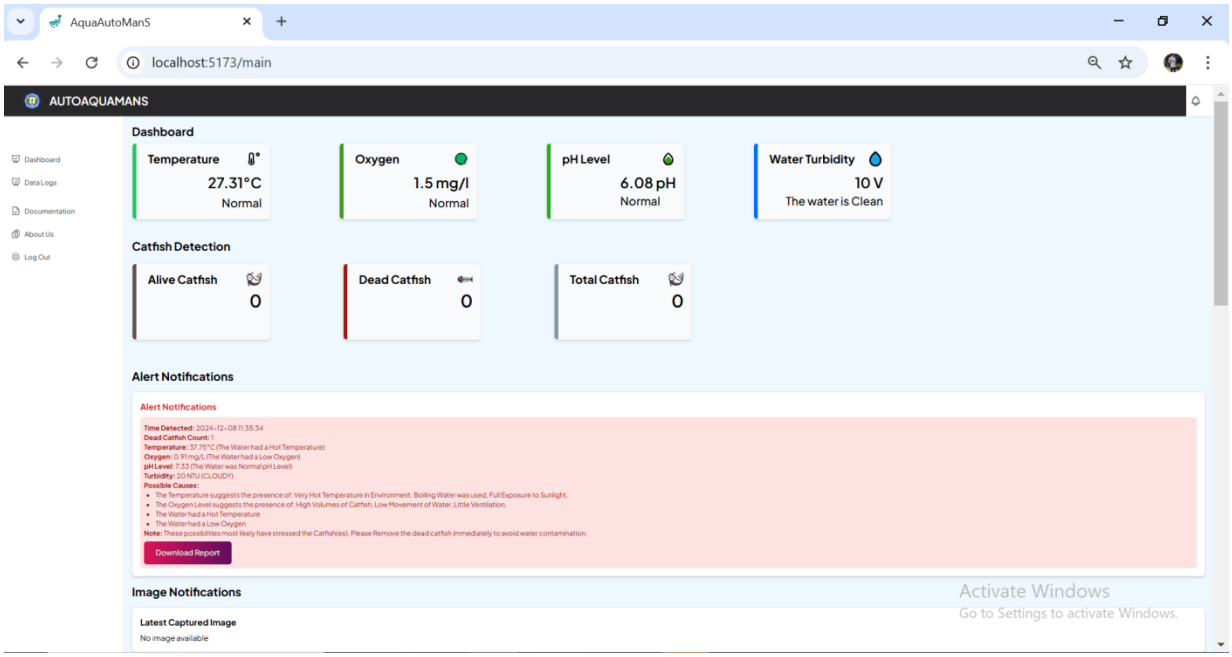


Figure 4.2 Website Dashboard

The system also gives the user the option to download the data gathered for each day any time the user desires to, or if a fluctuation happens. Once a catfish has died, the system will also notify the user that a catfish has died and will give the possible reasons as to why the catfish has died based on the data gathered. The user can also download the report that was generated by the system. The report includes the incident report, the tallied water parameters for the whole day, and the possible causes as to why the catfish has died. It will also recommend the user to remove the catfish immediately to avoid water contamination. Lastly, the users can also view the live monitoring of the actual catfish through the website.

4.3 Project Design



Figure 4.3.1 Project Model

Figure 4.3.1 shows the project model for the prototype of the Automated Aquaculture Monitoring System. Included here are the components used for collecting the data of the water aquarium such as the sensors which are used to detect any minor or major fluctuations of the environment of the water, and the webcam that is used to detect any dead or alive catfish that are present in the aquarium.

4.3.2 Project Model with Dimensions

Depicted here in the figure 4.3.2 is the 3D model of the prototype system for the Automated Catfish Aquaculture Monitoring system showcases a well-structured design, emphasizing both functionality and compactness. The main tank, measuring 91 cm in length, 45 cm in width, and 46.5 cm in height, provides ample space for monitoring catfish behavior and environmental conditions.

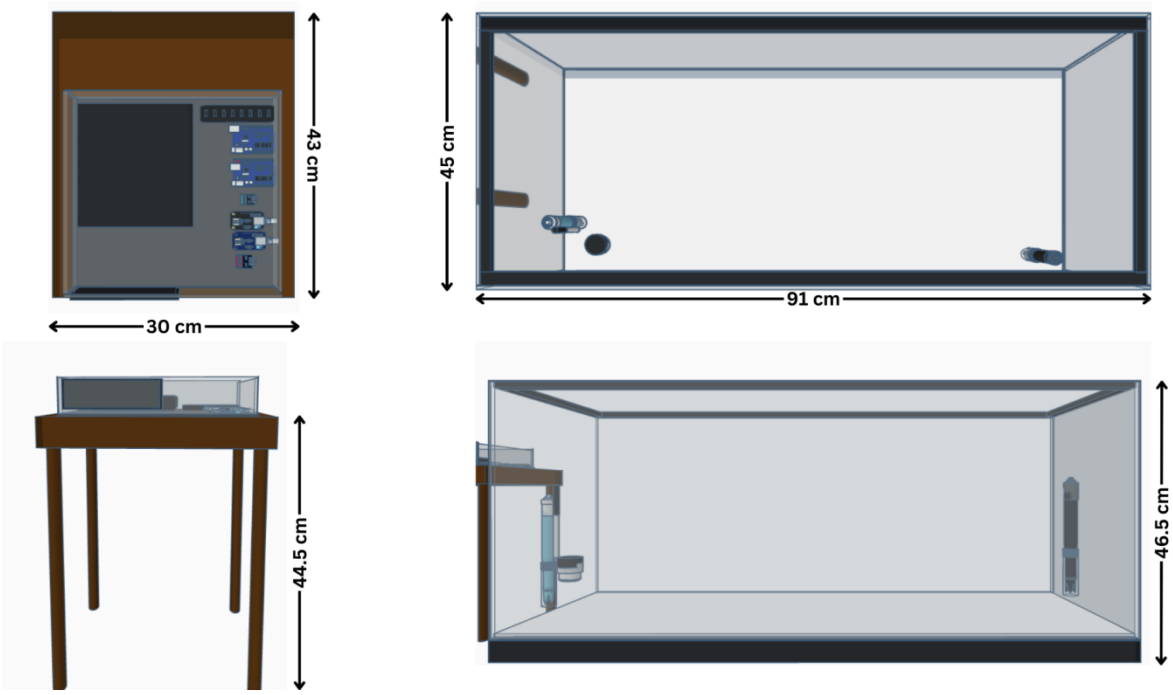


Figure 4.3.2 3D Model with Dimensions

The system integrates multiple specialized sensors throughout its framework, creating a comprehensive monitoring network that captures water quality metrics, flow patterns, and fish behavior in real time. At its core, the compact control unit (30 cm × 43 cm) houses advanced processing hardware and custom software that analyzes incoming data and allows for remote management via a secure interface. The thoughtful engineering extends to the physical design, with the entire apparatus standing 44.5 cm tall on a vibration-dampening stand that ensures stable operation even in busy aquaculture environments. Waterproof connections and quick-access panels reflect the designers' practical understanding of daily maintenance needs, while the modular architecture allows facilities to scale the system as operations grow. Early field tests have

demonstrated significant improvements in feeding efficiency and disease prevention, suggesting this technology could substantially increase yield while reducing resource consumption in both research and commercial fish farming applications.

4.3.3 3D Model Simulation

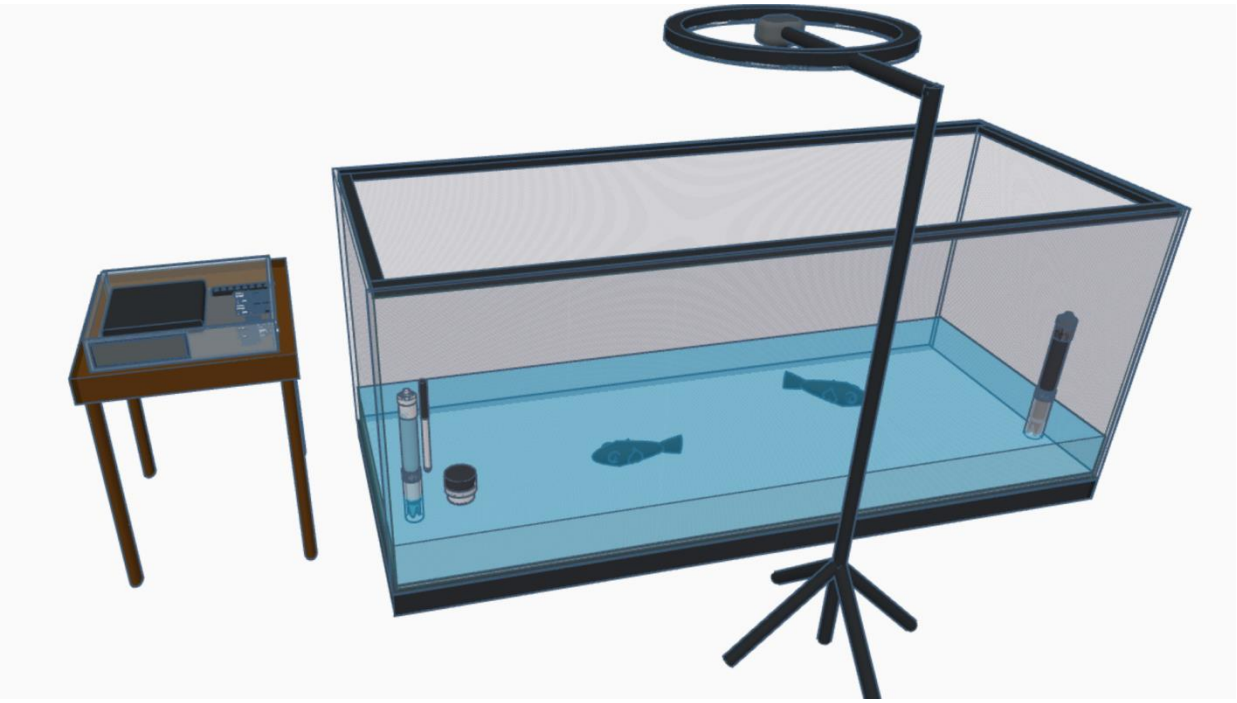


Figure 4.3.3 3D Model Simulation

Here in Figure 4.3.3, is the simulation of the Automated Catfish Aquaculture Monitoring system vividly illustrates the integration of technology within an aquaculture environment. The setup features a spacious tank, where catfish are monitored using an array of sensors strategically placed to capture vital data on water quality and fish behavior. The control unit, positioned on a nearby stand, serves as the central hub for

data processing and system management. Overhead, a lighting system ensures optimal visibility, enhancing the accuracy of monitoring. This simulation not only demonstrates the system's capability to maintain ideal conditions for catfish but also highlights the potential for real-time data analysis and decision-making. The design reflects a seamless blend of practicality and innovation, paving the way for advancements in sustainable aquaculture practices.

4.4 Concept of Operation

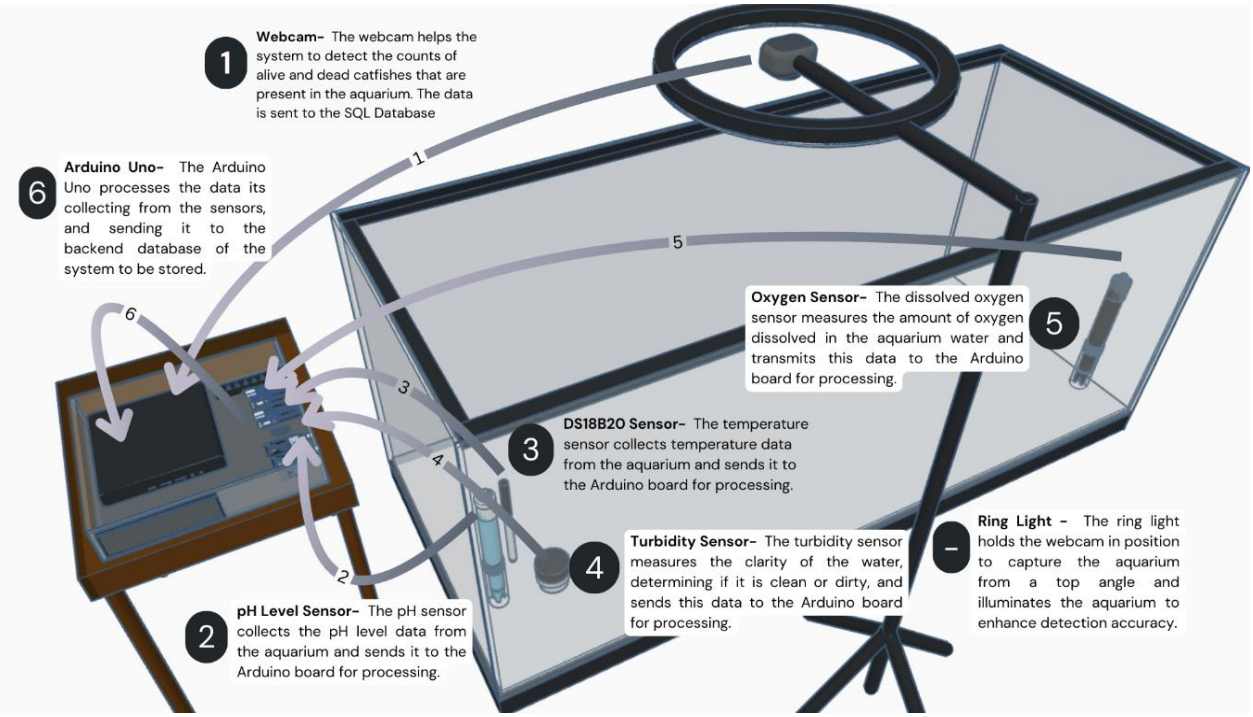


Figure 4.4 Concept of Operation of the System

Figure 4.4 shows the 3D prototype of the Automated Aquaculture Management System. This setup uses a Mini-PC as the central controller, while two Arduino Uno

boards, equipped with various sensor modules, act as data collectors. Each sensor monitors specific water conditions in the aquarium, capturing fluctuations in real-time. The data collected by the sensors is sent to the Arduinos, which then transmit it to a MySQL database for storage. This stored data can be accessed by a frontend website, allowing users to monitor the aquarium's environment effectively.

4.5 System Architecture

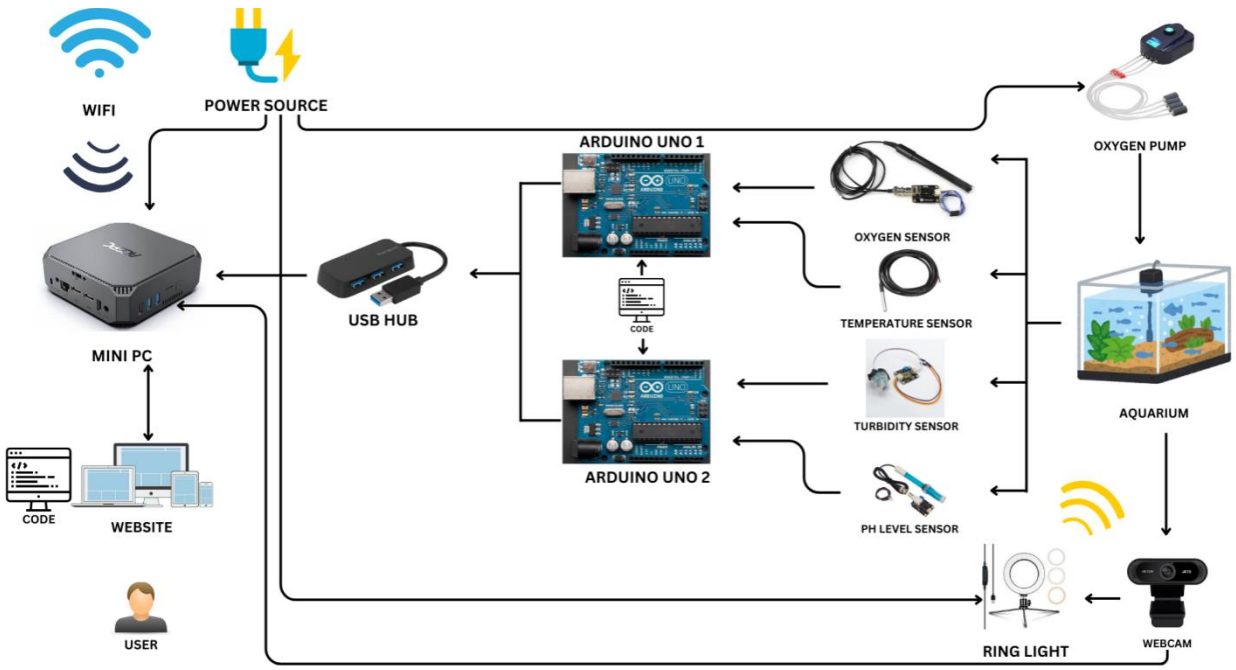


Figure 4.5.1 System Architecture

Figure 4.5.1 demonstrates the flow of the system architecture of the Automated Aquaculture Management System prototype. From each component of the prototype such as the sensors, the two Arduino Uno, mini-PC, along with the webcam and aquarium, all the way to its power source to give the sufficient power that the prototype needs. The

network used by the prototype is through Wi-Fi connection in order for the system to connect to the internet. Then the user can check through the website all of the live data that is needed to be monitored.

4.6 System Testing and Data Collection

The system testing phase was conducted in an aquarium to validate its performance. Sensors were deployed to measure water quality parameters such as temperature, pH, oxygen, and turbidity levels, with real-time data monitored via the integrated AI. Fish vitality detection was performed using image processing algorithms, which recorded fish movement, health status, and potential signs of stress.

- Temperature Monitoring: Data on temperature regulation and fluctuations.
- pH Levels: pH values tracked over different times of the day.
- Dissolved Oxygen (DO): Levels were analyzed to ensure fish had optimal oxygen conditions.
- Turbidity Levels: Used to analyze how clean or dirty the water is in the aquarium.
- Fish Vitality: Recorded through camera feeds and analyzed for abnormalities.

4.6.1 Analysis of Environmental Factors

For this section, it will show the comprehensive ranges of the water environment's changes, from the temperature, oxygen, pH level, and turbidity level, to properly understand where a catfish can survive and where the catfish may not survive. The analysis will include both optimal and threshold conditions for various catfish species, highlighting seasonal variations and the physiological adaptations that allow these

resilient fish to thrive in diverse aquatic habitats. Additionally, this examination will consider how these environmental parameters interact with one another, as the combined effect of multiple stressors can significantly impact catfish survival rates and overall population health in both natural and managed ecosystems.

Table 4.6.2: *Temperature Range for Catfish Survivability*

TEMPERATURE RANGE (°C)	CLASSIFICATION	EFFECTS ON CATFISH	CASE NO.
26-32	Normal Temperature Level	Optimal for catfish growth, feed intake and health will significantly increase.	Case 1
20-25, 33- 34	Minor Temperature Level	Minor stress and slightly reduces growth rate and feed intake. Health rates may also decrease.	Case 2
20-25, 33-34	Multiple Minor Water Fluctuation	Significant stress with reduced growth and feed intake; includes additional minor fluctuation like pH, and oxygen level changes.	Case 3
Below 19, Above 33	Critical Temperature Level	Rate of mortality significantly increases; catfish will possibly die in a few hours or days. May also include major water environmental fluctuations.	Case 4
Below 19, Above 33	Mortality Due to Temperature Changes	Significant temperature changes that lead to mortality of the catfish.	Case 5

As shown in Table 4.6.2, temperature regulation is vital for maintaining a healthy catfish aquaculture system. The normal temperature range (26–32°C) offers optimal conditions for catfish growth, feed intake, and overall health. Within this range, catfish thrive with maximum productivity. When temperatures fall into the minor fluctuation range (20–25°C, 27–32°C), there may be slight reductions in growth and health, but the impact is minimal, allowing catfish to continue developing. However, exposure to multiple minor

fluctuations (16–19°C, 33–35°C), such as combined temperature and pH level changes, can cause significant stress, weakening their immune system and slowing growth. In the presence of major fluctuations (below 16°C or above 35°C), catfish experience severe physiological stress, often resulting in high mortality rates due to extreme conditions, potentially exacerbated by other factors like extreme pH levels. Therefore, proper monitoring and control of water temperature and related environmental factors are essential to ensure catfish survival and productivity in aquaculture systems.

Table 4.6.3: *Dissolved Oxygen Range for Catfish Survivability*

DISSOLVED OXYGEN (mg/L)	CLASSIFICATION	EFFECTS ON CATFISH	CASE NO.
1.5 - 5	Normal Oxygen Level	Optimal for catfish growth, feed intake and health will significantly increase.	Case 1
1.0 - 1.4, 5.1 - 6	Stress Oxygen Range	Significant stress, reduced immunity, risk of disease.	Case 2
1.0 - 1.4, 5.1 - 6	Multiple Minor Fluctuations	Combined stress from oxygen and other factors like temperature or pH changes.	Case 3
Below 1, Above 7	Critical Oxygen Level	High mortality risk; catfish may die within hours.	Case 4
Below 1, Above 7	Mortality Due to Significant Oxygen Level Changes	Significant dissolved oxygen changes that lead to mortality of the catfish.	Case 5

As highlighted here in Table 4.6.3, dissolved oxygen levels play a significant role in catfish survival and growth. The normal range (1.5–5.0 mg/L) supports optimal metabolism, feeding efficiency, and overall health. Within the stress range (1.0–1.4 and

5.0–6.0 mg/L), catfish experience significant stress, leading to lethargy, increased disease susceptibility, and slower growth. When dissolved oxygen levels drop below 1.0 mg/L or exceed 7.0 mg/L, conditions become critical, posing a high mortality risk as catfish may suffocate within hours. Ensuring proper aeration and water circulation is essential to prevent oxygen depletion and maintain a healthy aquaculture environment.

Table 4.6.4: *pH Level Range for Catfish Survivability*

pH LEVEL	CLASSIFICATION	EFFECTS ON CATFISH	CASE NO.
6 - 7.5	Normal pH Level	Optimal for catfish growth, feed intake and health will significantly increase.	Case 1
5 - 5.9/ 7.6 - 8.5	Minor issues in pH Level	Minor stress conditions; reduced growth, feed intake, and immune function. The water could either have a slight acidic or alkaline pH level.	Case 2
5 - 5.9/ 7.6 - 8.5	Multiple Minor Water Fluctuation	Significant stress condition; that leads to lower feed efficiency and slow growth; includes additional minor fluctuation like temperature, and oxygen level changes.	Case 3
Below 5 - Above 8.5	Major pH and water environment fluctuation	Severe stress; catfish will possibly die in a few hours or days. The water could have a significant acidity or alkalinity level, and other water environmental fluctuation. May also include other major water environment fluctuations.	Case 4
Below 5 - Above 8.5	Mortality due to significant pH level changes	Significant pH level changes that lead to the mortality of the catfish	Case 5

Table 4.6.4 illustrates the significant impact of pH levels on catfish health and growth. The normal pH range (6.0–7.5) provides optimal conditions for metabolism, digestion, and immunity, promoting healthy growth. When the pH level shifts to slightly acidic or alkaline (5.0–5.9 or 7.6–8.5), catfish experience minor stress, which can lead to

reduced growth, feed intake, and immune function. These conditions may also involve slight temperature or oxygen level changes. In cases of multiple minor fluctuations, stress increases, further slowing growth and reducing feed efficiency. Extreme pH levels (below 4 or above 8.5) pose severe risks, causing significant stress, tissue damage, and potentially high mortality rates. Effective water quality management is essential to maintain a proper pH balance, ensuring sustainable catfish aquaculture.

4.7 PROCESS BLOCK DIAGRAM

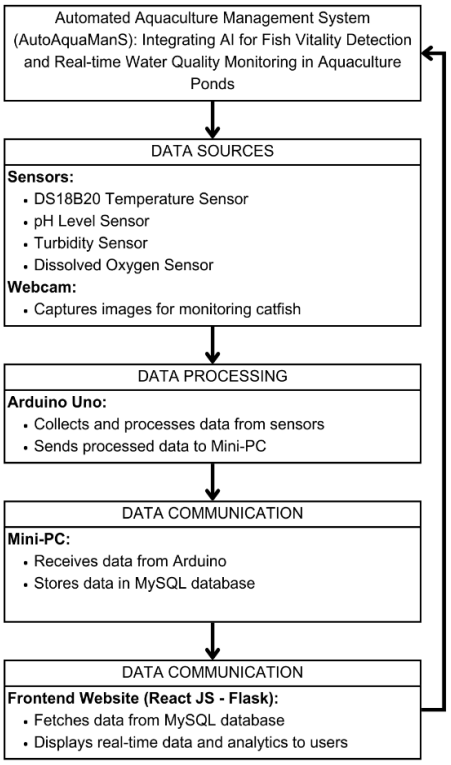


Figure 4.7.1 Process Block Diagram

As depicted in Figure 4.6.1, the Automated Aquaculture Management System is designed to seamlessly integrate various components to ensure efficient monitoring and

management of aquarium conditions. At the heart of the system are the data sources, which include a range of sensors and a webcam. These sensors covering temperature, pH, turbidity, and dissolved oxygen are crucial for capturing the real-time status of the aquarium environment. The webcam adds an additional layer by visually monitoring the catfish, providing insights into their behavior and health. Once the data is collected, it is processed by the Arduino Uno boards. These boards act as the system's intermediaries, taking raw data from the sensors and preparing it for further analysis.

The Arduinos ensure that the data is accurately captured and formatted before sending it to the Mini-PC.The Mini-PC plays a pivotal role in data communication. It receives the processed data from the Arduino boards and stores it in a MySQL database. This step is essential for maintaining a historical record of the aquarium's conditions, allowing for trend analysis and long-term monitoring. Finally, the user interface, presented through a frontend website, provides an accessible platform for users to interact with the system. By fetching data from the MySQL database, the website offers real-time updates and visualizations, enabling users to make informed decisions about the aquarium's management. This comprehensive approach ensures that all aspects of the aquarium environment are monitored and controlled, promoting the health and well-being of the aquatic life within.

4.8 TESTING AND EVALUATION

This section presents the results of testing the hardware, sensors, and web application components of the Automated Aquaculture Management System. The tests

ensure that all parts are functioning correctly and meet the expected performance standards.

Table 4.8.1: *Unit Testing of Sensors*

SENSORS	READINGS	DESCRIPTION	REMARKS	ACTION TAKEN
Temperature Sensor (DS18B20)	26°C	Measures the temperature of the water in the aquarium	The sensor was working properly	
Analog Dissolved Oxygen Sensor	4 mg/L	Measures the oxygen in the aquarium	The sensor displayed inaccurate readings	Properly calibrated to ensure accuracy
Analog pH Sensor/Meter Kit V2	Acidity / Alkalinity Level (0-14)	Measures the pH level of the aquarium	The sensor displayed inaccurate readings	Properly calibrated to ensure accuracy
Turbidity Sensor	10 NTU	Measures how clean the aquarium is	The sensor was working properly	

Table 4.8.1 shows the results of unit testing for the sensors used in the system, including the DS18B20 Temperature Sensor, Dissolved Oxygen Sensor, Analog pH Sensor, and Turbidity Sensor. For both Analog Dissolved Oxygen Sensor and Analog pH Sensor, both respective sensors did not display the proper results of each of its readings. The readings of each result were inaccurate based on the condition of the aquarium water. To fix the inaccuracy issue, the proponents calibrated both the Oxygen and pH sensors, and after that, the sensors displayed accurate readings of each data the sensors are reading.

Table 4.8.2: *Water Quality Condition Case Definitions*

Cases	Description	Parameters Affected	Expected Catfish Response	Remarks
Case 1	Normal Water Condition	No parameters are affected as the water condition is normal.	Healthy Growth	Conditions are optimal; maintain regular monitoring.
Case 2	One Minor Fluctuation	pH level, Temperature level, and Dissolved Oxygen	Minimal or no observable stress	Monitor to prevent further changes.

Case 3	Two or More Minor Fluctuations	Above or Below Average pH level, Temperature level, and Dissolved Oxygen levels	Stress range, reduced immunity	Address fluctuations promptly.
Case 4	Two or more Major Fluctuations	Extremely Low or High Level of pH level, Temperature level, and Dissolved Oxygen levels	High Risk of Mortality, need immediate intervention	Immediate intervention needed.
Case 5	Mortality Observed	Significant pH, Temperature, and Dissolved Oxygen levels that causes the Catfish to perish.	Catfish Death	Remove catfish to avoid further water contamination

Table 4.8.2 provides a comprehensive overview of various water quality conditions and their impact on catfish health, serving as a crucial tool for aquaculture management. In Case 1, optimal water conditions support healthy catfish growth, necessitating regular monitoring to maintain this state. Case 2 involves one minor fluctuation in parameters like a below or above average pH, temperature, dissolved oxygen, or turbidity levels, which may cause minimal to no stress, highlighting the need for close observation to prevent further changes. Case 3 describes scenarios with two or more minor fluctuations, placing catfish in a stress range that can reduce immunity, thus requiring prompt corrective actions. Case 4 presents a critical situation with major fluctuations, posing a high mortality risk and demanding immediate intervention to avert losses. Finally, Case 5 reflects severe conditions where significant deviations have led to catfish mortality, emphasizing the importance of investigating and rectifying underlying issues. This framework underscores the necessity of vigilant monitoring and timely responses to maintain optimal water conditions and ensure the health and survival of catfish. The proponents utilized the water level parameters provided in Section 4.6.1, which detail the ranges for temperature, dissolved oxygen, pH levels, and turbidity, to easily identify which case number is

affecting the health of the catfish. These parameters serve as a reference point for determining the specific conditions under which the catfish thrive or face stress, thereby guiding effective aquaculture management practices.

Water Quality Incident Report - Case 1

Report ID:	INC202412041528
Generated On:	2025-03-29 03:28:40
Generated By:	LexusBelisario
Case Level:	Case 1 - Normal
Description:	All parameters within optimal ranges.
Catfish Count:	Alive: 1.0 | Dead: 0.0

Current Readings

Temperature:	28.1
Status:	✓ Normal Temperature Level
Normal Range:	26-32°C
Effects:	Optimal for catfish growth, feed intake and health will significantly increase.

Oxygen:	2.1
Status:	✓ Normal Oxygen Level
Normal Range:	1.5-5.0 mg/L
Effects:	Optimal for catfish growth, feed intake and health will significantly increase.

Phlevel:	6.65
Status:	✓ Normal pH Level
Normal Range:	6.0-7.5
Effects:	Optimal for catfish growth, feed intake and health will significantly increase.

Recommendations

Priority:	Low
Action:	**Maintain Current Conditions**
Details:	**Current Status:** • Temperature (26-32°C): Optimal for catfish growth • Oxygen (1.5-5 mg/L): Optimal for feed intake • pH (6-7.5): Optimal for overall health **Regular Maintenance Actions:** • Check Sensors Regularly • Clean Aquarium Slightly • Maintain feeding schedule • Calibrate Sensors to ensure accurate readings • Monitor catfish behavior• Perform Weekly Cleaning of Aquarium and Sensors

Report generated on 2025-03-29 03:28:40 by LexusBelisario This report is automatically generated by the Aquaman Monitoring System

Figure 4.8.3 Case 1 Incident Report

As illustrated here in the Case 1 Incident Report, which is generated automatically by the system if it detects normal parameters. This means that the parameters are within optimal ranges, indicating a stable environment for catfish. Current readings show a temperature of 28.1°C, an oxygen level of 2.1 mg/L, and a pH level of 6.65, all of which fall within the ideal ranges for catfish growth and health. The recommendations emphasize maintaining these conditions with regular sensor checks, slight cleaning of the aquarium, and adherence to the feeding schedule. The priority is low, suggesting that the current environment is conducive to optimal catfish development and requires only routine maintenance to sustain these conditions.

Water Quality Incident Report - Case 2

Report ID:	INC202501310000
Generated On:	2025-01-31 00:00:00
Generated By:	LexusBelisario
Case Level:	Case 2 - Medium
Description:	Minor stress condition detected.
Catfish Count:	Alive: 1.0 | Dead: 0.0

Current Readings

Temperature:	26.6
Status:	Normal Temperature Level
Normal Range:	26-32°C
Effects:	Optimal for catfish growth, feed intake and health will significantly increase.

Oxygen:	2.97
Status:	Normal Oxygen Level
Normal Range:	1.5-5.0 mg/L
Effects:	Optimal for catfish growth, feed intake and health will significantly increase.

Phlevel:	5.5
Status:	Minor pH Level
Normal Range:	6.0-7.5
Effects:	Minor stress conditions; reduced growth, feed intake, and immune function.

Recommendations

Priority:	Medium
Action:	**Address Acidic pH Condition**
Details:	Minor acidic conditions detected Growth and immune impact possible **Actions required:** • Plan 25% water change • Test source water pH • Monitor pH every 2 hours • Check for acidic influences • Document pH changes

Report generated on 2025-01-31 00:00:00 by LexusBelisario This report is automatically generated by the Aquaman Monitoring System

Figure 4.8.4 Case 2 Incident Report

Here in Figure 4.8.4 is the system-generated Water Quality Incident Report for Case 2 which highlights a medium-priority issue concerning minor acidic conditions in the aquatic environment, with a pH level of 5.5, below the normal range of 6.0-7.5. This deviation poses potential risks to catfish growth, feed intake, and immune function, necessitating immediate corrective actions. Recommended measures include a 25% water change, testing the source water pH, and monitoring pH levels every two hours to mitigate stress conditions. By addressing these factors, the aquatic environment can be stabilized, ensuring optimal conditions for catfish health and growth.

Water Quality Incident Report - Case 3

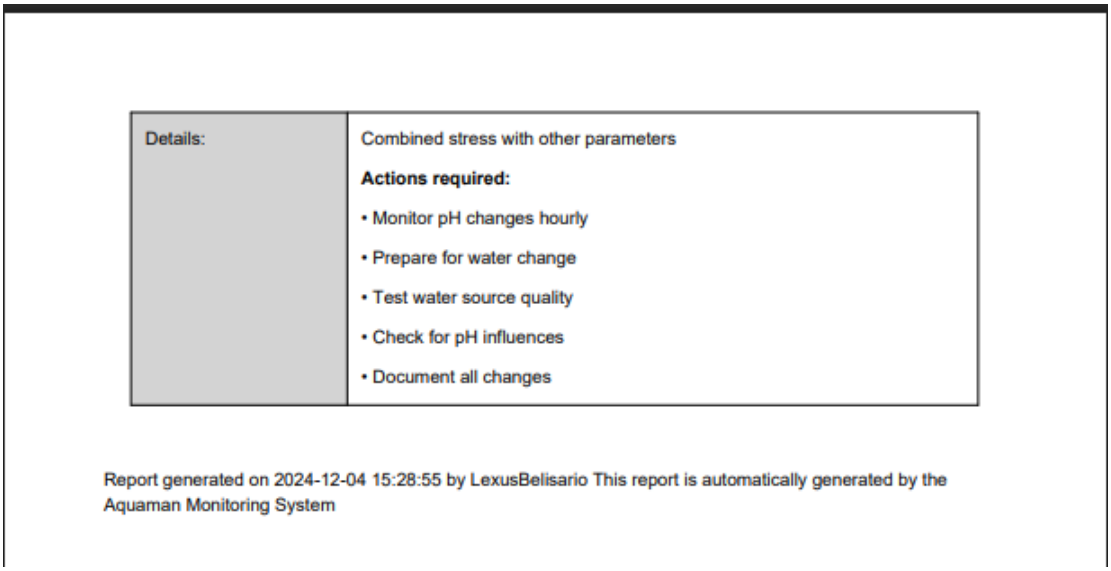
Report ID:	**INC202412041528**
Generated On:	**2024-12-04 15:28:55**
Generated By:	**LexusBelisario**
Case Level:	**Case 3 - High**
Description:	**Multiple stress conditions detected across parameters.**
Catfish Count:	**Alive: 1.0 | Dead: 0.0**

Current Readings

Temperature:	27.2
Status:	Normal Temperature Level
Normal Range:	26-32°C
Effects:	Optimal for catfish growth, feed intake and health will significantly increase.

Oxygen:	1.2
Status:	Stress Oxygen Range
Normal Range:	1.5-5.0 mg/L
Effects:	Significant stress, reduced immunity, risk of disease.

Phlevel:	5.65
Status:	Minor pH Level
Normal Range:	6.0-7.5
Effects:	Minor stress conditions; reduced growth, feed intake, and immune function.



Details:	Combined stress with other parameters **Actions required:** • Monitor pH changes hourly • Prepare for water change • Test water source quality • Check for pH influences • Document all changes

Report generated on 2024-12-04 15:28:55 by LexusBelisario This report is automatically generated by the Aquaman Monitoring System

Figure 4.8.5 Case 3 Incident Report

For Figure 4.8.5 is the actual Incident Report for Case 3, produced by the Aquamans Monitoring System, identifies a medium-priority issue due to minor acidic conditions, with a pH level of 5.5, below the optimal range of 6.0-7.5. This condition could negatively impact catfish growth, feed intake, and immune function. To address this, the report recommends a 25% water change, testing the source water pH, and monitoring pH levels every two hours to stabilize the environment. Additionally, the use of buffering agents is suggested to prevent future fluctuations, and a review of feeding practices is advised to ensure they are not contributing to the acidity. Regular maintenance of filtration and aeration systems is also recommended to maintain optimal conditions. These actions aim to ensure the health and growth of the catfish population by restoring and maintaining a balanced aquatic ecosystem.

Water Quality Incident Report - Case 4

Report ID:	**INC202411302018**
Generated On:	**2024-11-30 20:18:55**
Generated By:	**LexusBelisario**
Case Level:	**Case 4 - Critical**
Description:	**Critical levels detected in water parameters.**
Catfish Count:	**Alive: 1.0 | Dead: 0.0**

Current Readings

Temperature:	26.6
Status:	Normal Temperature Level
Normal Range:	26-32°C
Effects:	Optimal for catfish growth, feed intake and health will significantly increase.

Oxygen:	0.97
Status:	Critical Oxygen Level
Normal Range:	1.5-5.0 mg/L
Effects:	High mortality risk; catfish may die within hours.

Phlevel:	10.98
Status:	Critical pH Level
Normal Range:	6.0-7.5
Effects:	Severe stress; catfish will possibly die in a few hours or days.

Recommendations

Priority:	Critical
Action:	**CRITICAL PARAMETER ALERT**
Details:	IMMEDIATE ACTION REQUIRED Critical water parameters detected **Core actions required:** • Prepare for emergency water change • Monitor parameters every 30 minutes • Check all life support systems • Document all changes • Prepare emergency equipment

Priority:	Critical
Action:	**Critical Low Oxygen Alert**
Details:	Severe hypoxia condition **Emergency actions required:** • Maximize aeration immediately • Emergency water change • Add supplemental oxygen • Monitor fish continuously • Document oxygen changes

Priority:	Critical
Action:	**Critical High pH Alert**
Details:	Severe alkaline conditions **Emergency actions required:** • Immediate water change • Test source water • Check contamination sources • Monitor fish continuously • Document pH changes

Report generated on 2024-11-30 20:18:55 by LexusBelisario This report is automatically generated by the Aquaman Monitoring System

Figure 4.8.6 Case 4 Incident Report

For Figure 4.8.6 is the system-generated Water Quality Incident Report for Case 4, created by the Aquamans Monitoring System, indicates a critical situation with severe deviations in water parameters. The oxygen level is critically low at 0.97 mg/L, posing an immediate risk of high mortality for the catfish. Additionally, the pH level is critically high at 10.98, causing severe stress. Immediate actions are required, including emergency water changes, maximizing aeration, and adding supplemental oxygen. Continuous monitoring and documentation of changes are essential to stabilize conditions and prevent potential fatalities.

Case 5 - Dead Catfish Detected

IMMEDIATE ACTION REQUIRED: Remove dead catfish immediately to prevent water contamination!

Water Quality Regulation Required:

• CRITICAL: Water temperature is dangerously high. Take immediate action: - Cool down water temperature immediately - Add fresh, cooler water - Provide shade from direct sunlight - Consider adding cooling system

• Low oxygen levels detected: - Increase water movement - Check aeration system - Monitor oxygen levels hourly

Required Immediate Actions:

1. Remove dead catfish IMMEDIATELY regardless of water parameters
2. Document time and conditions of death
3. Perform water quality tests
4. Implement recommended water regulations as listed above
5. Monitor remaining catfish for signs of stress
6. Test water parameters every 2-3 hours after changes

Incident Summary

Time of Death	Total Catfish	Dead Catfish	Mortality Rate
2024-12-08 11:35:34	0.0	1.0	100.00%

Water Parameters at Time of Death

Parameter	Value	Status	Normal Range
Temperature	37.75°C	Hot	26-32°C
Oxygen	0.91 mg/L	Low	1.5-5.0 mg/L
pH Level	7.33	Normal	6.0-7.5

Figure 4.8.7 Case 5 Incident Report

In Figure 4.8.7 is the system-generated report for Case 5 indicates a critical situation with the detection of a dead catfish, necessitating immediate action to prevent water contamination. The water temperature is dangerously high at 37.75°C, and oxygen levels are critically low at 0.91 mg/L. Immediate actions include removing the dead catfish, cooling the water, increasing aeration, and monitoring oxygen levels hourly. It is crucial to document the conditions of death, perform water quality tests, and implement recommended regulations to stabilize the environment and prevent further stress on the remaining catfish.

Dead Catfish Documentation

Image captured at: 2024-12-08 11:32:32



Figure 4.8.8 Dead Catfish Image Report

Here in Figure 4.8.8 is the Dead Catfish Documentation captured on 2024-12-08 at 11:32:32 a part of the Case 5 Incident Report that provides visual evidence of the incident reported in Case 5. The image clearly shows the deceased catfish, highlighting the urgency of addressing the critical water conditions that led to this outcome. Immediate actions, as outlined in the report, are essential to prevent further contamination and ensure the health of any remaining aquatic life. This documentation supports the need for rapid intervention and ongoing monitoring to stabilize the environment.

Table 4.8.3: *Integration Testing for Sensor Data Processing and Frontend Display*

Trial Case No.	Component(s) Involved	Trial Scenario	Expected Result	Actual Result	Remarks	Action Taken
Trial 1	Temperature Sensor → Backend → Database → Dashboard	Verify temperature sensor readings are processed database and frontend correctly.	The temperature value should be correctly displayed on the dashboard.	Sensor failed to send data to backend; value not displayed on dashboard.	There is an error in the line of code for the connection of sensor to database to website	Fixed the SQL code in the Python backend
Trial 2	pH Sensor → Backend → Database → Dashboard	Verify pH sensor readings are stored and displayed in the database and frontend correctly.	The pH value should be stored in the database and visible on the dashboard.	pH sensor data displayed correctly.	The sensor showed accurate results in the dashboard	
Trial 3	Oxygen Sensor → Backend → Database → Dashboard	Ensure oxygen level readings are correctly recorded and displayed in the database and frontend.	Oxygen levels should be correctly displayed on the dashboard	Oxygen sensor data	The sensor showed accurate results in the dashboard	
Trial 4	Turbidity Sensor → Backend → Database → Dashboard	Ensure turbidity level readings are stored and displayed in the database and frontend correctly.	The turbidity value should be stored and displayed accurately.	Turbidity sensor data displayed correctly.	The sensor showed accurate results in the dashboard	

Table 4.8.3 outlines the results of integration testing for sensor data processing and frontend display, focusing on the seamless flow of data from sensors through the backend to the dashboard. The testing aimed to ensure that data collected by various sensors is accurately processed, stored, and displayed, providing real-time insights into the aquarium's environmental conditions.

In Trial 1, the temperature sensor was tested to verify that its readings were correctly processed through the backend, stored in the database, and displayed on the dashboard. However, the sensor failed to send data to the backend, resulting in the temperature value not being displayed. This issue was traced to an error in the SQL code responsible for connecting the sensor to the database and website, which was subsequently fixed in the Python backend.

Trial 2 focused on the pH sensor, testing the correct storage and display of its readings. The pH sensor data was displayed correctly on the dashboard, indicating successful integration without any issues. Similarly, Trial 3 tested the oxygen sensor to ensure that oxygen level readings were accurately recorded and displayed. The oxygen sensor data was displayed correctly, confirming the system's reliability in processing oxygen data.

Finally, Trial 4 verified the correct storage and display of turbidity level readings. The turbidity sensor data was displayed accurately, demonstrating successful integration. Overall, the integration testing highlighted a critical issue with the temperature sensor's data processing, which was promptly addressed. The successful integration of the pH,

oxygen, and turbidity sensors demonstrates the system's capability to handle and display sensor data effectively, ensuring accurate monitoring of the aquarium environment.

Table 4.8.4: *First Week Data Collection*

Date	Water Environment	Sensor Readings	Case No.	Catfish Reports	Recommendations
Jan 18	High Oxygen, Acidic Water	**Temperature**: 26.43°C **Oxygen**: 6.32 mg/L **pH Sensor**: 3.60 **Turbidity**: 10 NTU	Case 4	Stress range; potential reduced immunity and risk of disease.	Adjust pH Level and Decrease Oxygen Level
Jan 19	High Oxygen, Acidic Water	**Temperature**: 25.47°C **Oxygen**: 6.26 mg/L **pH Sensor**: 4.29 **Turbidity**: 10 NTU	Case 4	Stress range; potential reduced immunity and risk of disease.	Adjust pH Level and Decrease Oxygen Level
Jan 20	High Oxygen, Acidic Water	**Temperature**: 26.25°C **Oxygen**: 5.51 mg/L **pH Sensor**: 3.71 **Turbidity**: 14.66 NTU	Case 4	Stress range; potential reduced immunity and risk of disease.	Adjust pH Level and Decrease Oxygen Level
Jan 21	Below, Average Temperature, High Oxygen, Acidic Water	**Temperature**: 25.69°C **Oxygen**: 6.20 mg/L **pH Sensor**: 2.02 **Turbidity**: 20 NTU	Case 4	Mild stress; reduced growth and feed efficiency.	Slightly Increase temperature, adjust pH Level and Decrease Oxygen Level
Jan 22	Below, Average Temperature, High Oxygen, Acidic Water	**Temperature**: 25.47°C **Oxygen**: 6.32 mg/L **pH Sensor**: 4.15 **Turbidity**: 25 NTU	Case 4	Mild stress; reduced growth and feed efficiency.	Slightly Increase temperature, adjust pH Level and Decrease Oxygen Level
Jan 23	Below, Average Temperature, High Oxygen, Acidic Water	**Temperature**: 24.98°C **Oxygen**: 6.39 mg/L **pH Sensor**: 4.10 **Turbidity**: 25 NTU	Case 4	Stress range; potential reduced immunity and risk of disease.	Slightly Increase temperature, adjust pH Level and Decrease Oxygen Level
Jan 24	Below, Average Temperature, High Oxygen, Acidic Water	**Temperature**: 25.36°C **Oxygen**: 6.38 mg/L **pH Sensor**: 4.09 **Turbidity**: 20 NTU	Case 4	Stress range; potential reduced immunity and risk of disease.	Change the Water, and adjust the temperature and pH level of the Water.

As shown here in Table 4.8.4, during the data gathering period, the aquarium experienced various conditions. Initially, the water had high oxygen and was acidic, with temperatures ranging from 26.25°C to 26.59°C and pH levels between 2.02 and 4.29. Turbidity varied, peaking at 25 NTU. These conditions were stressful for the catfish, classified under Case 4, indicating potential reduced immunity and disease risk. Recommendations included adjusting pH levels and decreasing oxygen levels. On January 24th, a water change was advised to improve clarity and stabilize conditions. The

team followed these recommendations by using buffering agents, adjusting temperature, and performing water changes as needed. These actions effectively maintained the catfish's health, with no mortality reported despite the challenging conditions.

Table 4.8.5: *Second Week Data Collection*

Date	Water Environment	Sensor Readings	Case No.	Catfish Reports	Recommendations
Jan 25	Below Average Temperature, Acidic Water	**Temperature**: 25.41°C **Oxygen**: 5.18 mg/L **pH Sensor**: 4.55 **Turbidity**: 25 NTU	Case 4	Stressful conditions; reduced growth and immune function.	Stabilize pH level and slightly improve the temperature
Jan 26	Very Acidic Water, Low Oxygen, Cloudy Water	**Temperature**: 26.05°C **Oxygen**: 3.51 mg/L **pH Sensor**: 4.14 **Turbidity**: 25 NTU	Case 4	High mortality risk; immediate intervention needed.	Immediate pH and oxygen correction needed.
Jan 27	Below Average Temperature, Very Acidic Water, High Oxygen, Cloudy Water	**Temperature**: 24.12°C **Oxygen**: 2.36 mg/L **pH Sensor**: 2.99 **Turbidity**: 32 NTU	Case 4	High mortality risk; immediate intervention needed.	Adjust the pH level and improve water clarity and temperature.
Jan 28	Below Average Temperature, Very Acidic Water, Cloudy Water	**Temperature**: 29.98°C **Oxygen**: 4.91 mg/L **pH Sensor**: 4.60 **Turbidity**: 10 NTU	Case 4	High mortality risk; immediate intervention needed.	Adjust the pH level and improve water clarity and temperature.
Jan 29	Very Acidic Water, Cloudy Water	**Temperature**: 26.60°C **Oxygen**: 2.97 mg/L **pH Sensor**: 4.50 **Turbidity**: 10 NTU	Case 4	High mortality risk; immediate intervention needed.	Improve the pH level and change the water of the aquarium
Jan 30	Very Acidic Water, Cloudy Water	**Temperature**: 29.07°C **Oxygen**: 4.52 mg/L **pH Sensor**: 4.64 **Turbidity**: 10 NTU	Case 4	High mortality risk; immediate intervention needed.	Improve the pH level and change the water of the aquarium
Jan 31	Very Acidic Water, Cloudy Water	**Temperature**: 27.29°C **Oxygen**: 1.48 mg/L **pH Sensor**: 6.12 **Turbidity**: 10 NTU	Case 4	High mortality risk; immediate intervention needed.	Improve the pH level and change the water of the aquarium

According to Table 4.8.5, during the second week, the aquarium faced challenging conditions with very acidic and cloudy water. Temperatures ranged from 24.12°C to 29.98°C, oxygen levels varied between 1.48 mg/L and 6.12 mg/L, and pH levels were consistently low, from 4.14 to 4.64. Turbidity was high, peaking at 32 NTU. These conditions were classified under Case 4, indicating a high mortality risk and the need for immediate intervention. Recommendations included stabilizing the pH, improving water

clarity, and correcting oxygen levels. The team responded by performing water changes to reduce turbidity and acidity, adjusting the pH with buffering agents, and monitoring oxygen levels. Despite the high-stress environment, these proactive measures, guided by the system's recommendations, helped maintain the catfish's health, with no mortality reported. Timely interventions were crucial in mitigating the risks associated with the adverse water conditions.

Table 4.8.6: *Third Week Data Collection*

Date	Water Environment	Sensor Readings	Case No.	Catfish Reports	Recommendations
Feb 1	Normal Water Environment	**Temperature**: 26.05°C **Oxygen**: 2.66 mg/L **pH Sensor**: 7.54 **Turbidity**: 15 NTU	Case 1	The catfish is healthy and thriving under stable conditions.	Maintain Water Condition
Feb 2	High pH Level, Below Average Temperature.	**Temperature**: 25.82°C **Oxygen**: 2.23mg/L **pH Sensor**: 8.44 **Turbidity**: 16 NTU	Case 3	Stressful conditions; reduced growth and immune function.	Lower pH to neutral and increase temperature.
Feb 3	Very High Oxygen, Below Average Temperature	**Temperature**: 25.16°C **Oxygen**: 6.71 mg/L **pH Sensor**: 7.11 **Turbidity**: 15 NTU	Case 4	High mortality risk; immediate intervention needed.	Monitor oxygen levels and maintain temperature.
Feb 4	Acidic Level of pH level	**Temperature**: 26.3°C **Oxygen**: 2.08 mg/L **pH Sensor**: 4.31 **Turbidity**: 15 NTU	Case 4	High mortality risk;Immediate intervention needed.	Correct pH to neutral.
Feb 5	Acidic Level of Water, Below Average Temperature Level	**Temperature**: 25°C **Oxygen**: 2.53 mg/L **pH Sensor**: 4.82 **Turbidity**: 15 NTU	Case 3	Stressful conditions; reduced growth and immune function.	Stabilize pH and temperature.
Feb 6	Below Average Temperature, Acidic Range of Water	**Temperature**: 25.12 °C **Oxygen**: 4.10 mg/L **pH Sensor**: 6.45 **Turbidity**: 15 NTU	Case 4	High mortality risk; immediate intervention needed.	Maintain pH and increase temperature.
Feb 7	Normal Water Environment	**Temperature**: 27°C **Oxygen**: 1.27 mg/L **pH Sensor**: 6.94 **Turbidity**: 15 NTU	Case 1	The catfish is healthy and stable under normal conditions.	Maintain Water Condition

During the third week, as detailed in Table 4.8.6, the aquarium conditions varied significantly, with both stable and challenging environments observed. The week began and ended with normal water conditions, as indicated by sensor readings on February 1st

and 7th, where temperatures were around 26.05°C to 27°C, oxygen levels were 2.66 mg/L and 1.27 mg/L, and pH levels were 7.54 and 6.94, respectively. These conditions were classified under Case 1, with the catfish thriving and no interventions needed beyond maintaining the current water conditions. However, mid-week presented more challenging scenarios. On February 2nd, a high pH level and below-average temperature were recorded, leading to stressful conditions for the catfish. The system recommended lowering the pH to neutral and increasing the temperature. On February 3rd, very high oxygen levels combined with below-average temperatures posed a high mortality risk, prompting the system to advise monitoring oxygen levels closely and maintaining temperature. Further fluctuations were noted on February 4th and 5th, with acidic water conditions and below-average temperatures.

The system classified these under Case 3 and Case 4, recommending pH stabilization and temperature adjustments. The proponents responded by adjusting the pH with buffering agents and using heaters to increase the temperature, as suggested by the system. They diligently monitored the conditions, ensuring that any deviations were promptly addressed. Despite these fluctuations, the catfish remained healthy, with no mortality reported. For this week, the proponents took an immediate action based on what the system is recommending, in order to address and regulate the water fluctuations, such as stabilizing and regulating the pH level of the water, removing anything that can cause

a high level of oxygen, and further maintenance of the aquarium by cleaning it and changing the water of the aquarium to avoid polluting the water of the aquarium.

Table 4.8.8: *Fourth Week Data Collection*

Date	Water Environment	Sensor Readings	Case No.	Catfish Reports	Recommendations
Feb 8	Below Average Temperature, High Oxygen	**Temperature**: 25.46°C **Oxygen**: 3.17 mg/L **pH Sensor**: 5.61 **Turbidity**: 15 NTU	Case 2	Mild stress; reduced growth and feed efficiency.	Adjust pH and monitor oxygen.
Feb 9	Below Average Temperature, Acidic Range of Water	**Temperature**: 25.62°C **Oxygen**: 2.97 mg/L **pH Sensor**: 6.34 **Turbidity**: 25 NTU	Case 3	Stressful conditions; reduced growth and immune function.	Improve water clarity and stabilize pH.
Feb 10	Below average temperature range of water	**Temperature**: 25.10°C **Oxygen**: 2.57 mg/L **pH Sensor**: 7.69 **Turbidity**: 18 NTU	Case 2	Acceptable temperature; slight reduction in growth and health.	Maintain temperature and monitor pH.
Feb 11	Below Average Temperature, High Oxygen	**Temperature**: 25.62°C **Oxygen**: 2.97 mg/L **pH Sensor**: 7.43 **Turbidity**: 15 NTU	Case 4	High mortality risk; immediate intervention needed.	Monitor oxygen and maintain temperature.
Feb 12	Below Average Temperature, High Oxygen	**Temperature**: 25.90 °C **Oxygen**: 6.65 mg/L **pH Sensor**: 7.24 **Turbidity**: 10 NTU	Case 4	High mortality risk; immediate intervention needed.	Monitor oxygen levels.
Feb 13	Below Average Temperature, Low Oxygen	**Temperature**: 25.61 °C **Oxygen**: 1.25 mg/L **pH Sensor**: 7.52 **Turbidity**: 10 NTU	Case 2	Mild stress; reduced growth and feed efficiency.	Immediate oxygenation needed.
Feb 14	Below Average Temperature, Low Oxygen	**Temperature**: 25.73 °C **Oxygen**: 1.44 mg/L **pH Sensor**: 6.75 **Turbidity**: 10 NTU	Case 3	Stressful conditions; reduced growth and immune function.	Increase oxygen levels.

In the fourth week, as outlined in Table 4.7.8, the aquarium conditions were predominantly characterized by below-average temperatures and fluctuating oxygen levels. The sensor readings indicated temperatures ranging from 25.10°C to 25.90°C, with oxygen levels varying significantly from 1.25 mg/L to 6.65 mg/L. The pH levels were generally stable, ranging from 5.61 to 7.69, while turbidity varied between 10 NTU and 25 NTU. The system identified these conditions as causing mild to high stress for the catfish,

with classifications under Case 2, Case 3, and Case 4. On February 8th and 9th, the system recommended adjusting the pH and monitoring oxygen levels due to mild stress and reduced growth efficiency. As the week progressed, high oxygen levels on February 11th and 12th posed a high mortality risk, prompting the system to advise close monitoring of oxygen and maintaining temperature stability. Towards the end of the week, low oxygen levels on February 13th and 14th led to mild stress, with the system suggesting immediate oxygenation to improve conditions.

The proponents responded by adjusting the pH, performing water changes to improve clarity, and using aeration techniques to regulate oxygen levels, as recommended by the system. Despite the fluctuating conditions, the catfish remained healthy, with no mortality reported. The proponents' diligent monitoring and timely interventions, guided by the system's recommendations, were essential in maintaining a stable environment and supporting the catfish's resilience throughout the week.

4.8.9 Weekly Data Overview

The Weekly Data Overview Table provides a detailed comparison of the environmental conditions and interventions applied over a four-week period in the catfish aquarium. As shown in Table 4.8.9, In the first week, the predominant conditions were characterized by below-average temperatures and acidic water, with tallied sensor readings showing an average temperature of 25.5°C, oxygen at 1.3 mg/L, pH at 5.0, and turbidity at 15.7 NTU. As a result, the catfish exhibited moderate stress, with visible

peeling of the skin, but no mortality occurred. The system suggested adjusting the pH level by changing the water, which the proponents implemented effectively.

Table 4.8.9: *Weekly Data Overview*

WEEK	PREDOMINANT WATER CONDITIONS	TALLIED SENSOR READINGS	ACTION TAKEN	CATFISH CONDITION
1	Below Average Temperature, High Oxygen, Acidic Water	**Temperature**: 25.47°C **Oxygen**: 6.02 mg/L **pH Sensor**: 3.71 **Turbidity**: 20 NTU	Adjusted pH level and slightly increased temperature.	Stress range; potential reduced immunity and risk of disease. No mortality.
2	Very Acidic Water, Cloudy Water	**Temperature**: 26.60°C **Oxygen**: 2.97 mg/L **pH Sensor**: 4.50 **Turbidity**: 25 NTU	Immediate Water Change, to clean the water and for pH correction	High Stress; skin of the catfish visibly peeled. No mortality
3	Below Average Temperature, Fluctuating pH and Oxygen Levels	**Temperature**: 25.8 °C **Oxygen**: 3.0 mg/L **pH Sensor**: 6.1 **Turbidity**: 15.7 NTU	Turned off the oxygen pump to regulate oxygen.	Moderate Stress; No mortality
4	Below Average Temperature, Fluctuating Oxygen Level	**Temperature**: 25.6 °C **Oxygen**: 3.0 mg/L **pH Sensor**: 6.8 **Turbidity**: 15.3 NTU	Monitored the oxygen.	Mild Stress; No mortality

During the second week, the conditions worsened to very acidic and cloudy water, with sensor readings indicating a temperature of 25.9°C, oxygen at 2.8 mg/L, pH at 3.9, and turbidity at 27.1 NTU. The catfish experienced high stress, again with visible skin peeling, but no mortality was observed. The system recommended an immediate water change to clean the water and correct the pH, which the proponents promptly executed. The third week presented below-average temperatures with fluctuating pH and oxygen levels. The sensor readings averaged a temperature of 25.8°C, oxygen at 3.0 mg/L, pH at 6.1, and turbidity at 15.7 NTU. The catfish were under moderate stress, but no mortality

occurred. The system advised turning off the oxygen pump to regulate oxygen levels, a recommendation that the proponents followed to stabilize the environment. In the fourth week, the aquarium experienced below-average temperatures and fluctuating oxygen levels, with sensor readings showing a temperature of 25.6°C, oxygen at 3.0 mg/L, pH at 6.8, and turbidity at 15.3 NTU.

The catfish were under mild stress, with no mortality reported. The system suggested monitoring the oxygen levels closely, which the proponents did to ensure the health of the catfish. The results of the tallied sensor readings of the weekly findings were meticulously calculated by the proponents, who averaged the daily readings to provide a comprehensive overview of each week's conditions. This approach allowed them to identify trends and implement timely interventions, guided by the system's recommendations, to maintain a conducive environment for the catfish.

4.9 SCHEMATIC DIAGRAM

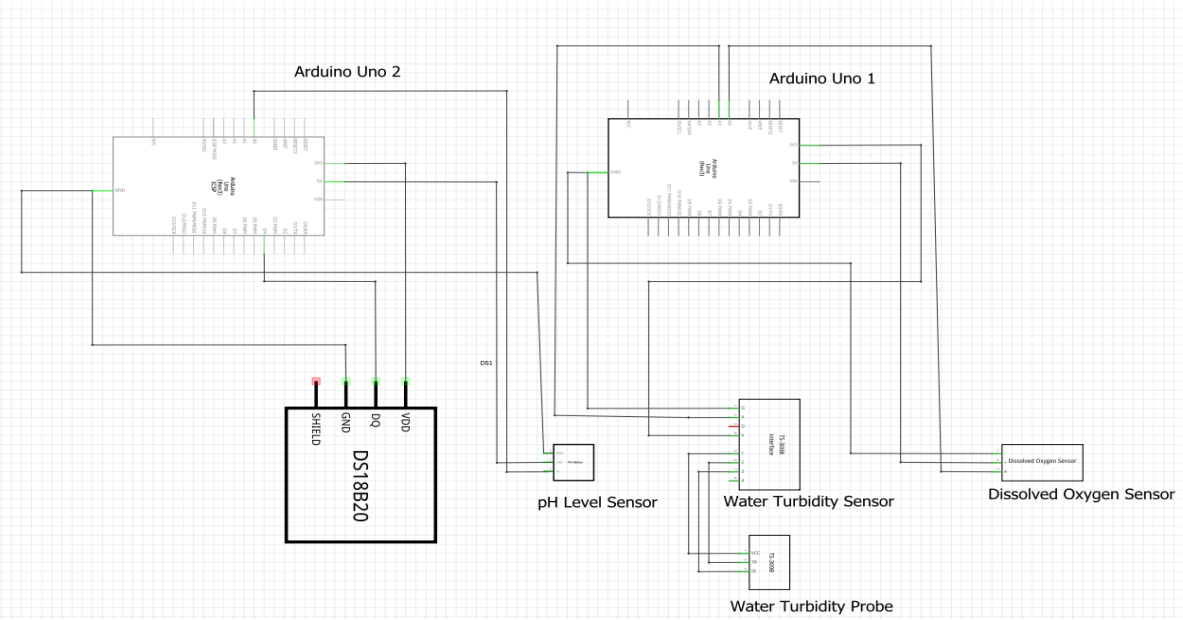


Figure 4.9.0 Schematic Diagram

Figure 4.9 illustrates the wiring schematic for the Automated Aquaculture Management System, highlighting the integration of multiple sensors with two Arduino Uno boards. Arduino Uno 1 is connected to the pH level sensor, water turbidity sensor, and dissolved oxygen sensor, managing the data collection from these critical environmental parameters. Meanwhile, Arduino Uno 2 is dedicated to the DS18B20 temperature sensor, ensuring precise temperature monitoring. This setup allows for efficient data processing and communication, with each Arduino handling specific tasks to optimize system performance. The clear separation of sensor responsibilities between the two Arduinos enhances reliability and ensures accurate data collection for maintaining optimal aquarium conditions.

4.10 AI Model

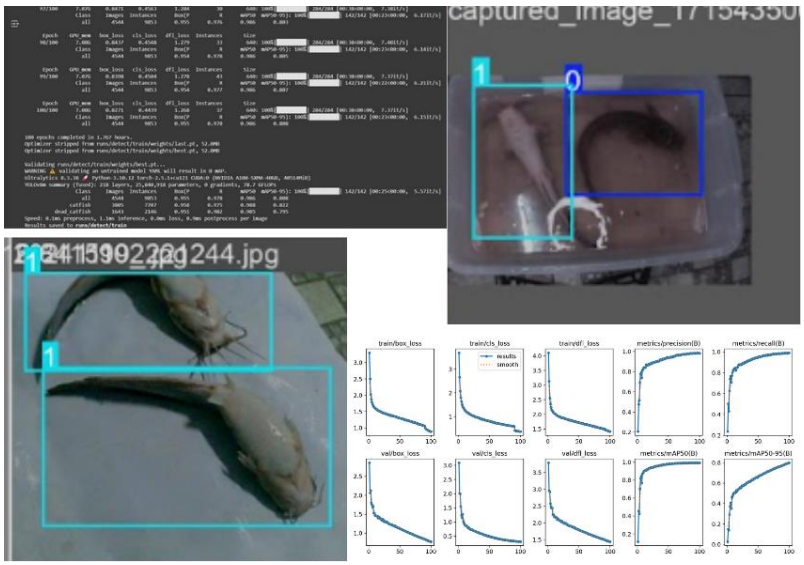


Figure 4.10 AI Model

The provided images demonstrate the results of a custom object detection model trained to identify two classes of fish. In the first and second images, the model successfully detects and classifies fish using bounding boxes and class labels. The class labels "0" and "1" correspond to two distinct fish types, and the model accurately identifies multiple instances in varied environments—one on a dry surface and the other submerged in water. This indicates that the model performs reliably across different backgrounds and lighting conditions, highlighting its robustness and generalization capabilities.

The training performance chart visualizes the model's learning progress over 100 epochs. Both training and validation losses, including box loss, classification loss, and distribution focal loss, show a consistent downward trend, which suggests effective learning and convergence without signs of overfitting. Precision and recall metrics

improve steadily, approaching values close to 1.0, while the mean Average Precision (mAP) at 0.5 and 0.5:0.95 thresholds also increase, reaching approximately 0.9 and 0.8 respectively. These results indicate that the model achieves high accuracy in object localization and classification, demonstrating its suitability for practical deployment in fish detection tasks.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

5.1 Summary

The purpose of the Automated Aquaculture Monitoring System is to help catfish keepers to properly determine how a catfish can die inside the aquarium that is caused by environmental changes of the water. The project is designed to help catfish keepers to help determine what kind of water environmental changes occur overtime, preventing the catfish from easily perishing every time the water changes its temperature level, oxygen level, pH level, and turbidity level. The proponents used Python-Flask as the backend language for the development of this project, React JavaScript for the Frontend, XAMPP to create a local server and store the data that the system collects, C++ was also used for the proper code for the Arduino UNO in order for the system to properly receive the data that the sensors are receiving. Lastly for the detection, the proponents used YoloV8 to train the proper data sets, and properly detect the number and the status of each catfish.

The project system's objective was met as it can determine if a catfish is alive or dead along with the possible causes of a death of a catfish if it dies using the data that every sensor receives and stores, it can also count how many catfish are present in the aquarium but limited to 5, and lastly, the system can also be used everywhere the user go as long as they have an internet connection, they can properly manage and download

the necessary data that is needed to check if a fluctuation in the environment of the aquarium occurs, or if a catfish has died.

5.2 Conclusion

After conducting all the research, the proponents came to the conclusion that the Automated Aquaculture Monitoring System had successfully met the study's goal and demonstrated its effectiveness and efficiency when implemented in catfish farms, business owners and fish keepers. Therefore, the proponents concluded that;

1. The development of the Automated Aquaculture Monitoring System tackled the challenges initially faced head-on. The proponents found that accurate monitoring and data display could be achieved through carefully selected sensors. They installed pH level, turbidity, dissolved oxygen, and temperature sensors that worked together to capture the essential environmental conditions in the water. All this information was stored locally on a Mini PC, while fish keepers could easily check current conditions through visual dashboards. These practical features gave hobbyists a reliable way to keep tabs on their African Catfish without constant manual checking.
2. Looking at long-term benefits, the system proved to be more than just a data collector. It became a watchful guardian that never sleeps. By continuously tracking water conditions over weeks and months, patterns emerged that helped prevent problems before they started. When something wasn't right maybe the temperature suddenly spiked or oxygen levels dropped the system immediately

sent alerts to the user's device. This quick notification system meant fish keepers could fix issues before their catfish suffered, saving both fish lives and the keeper's investment in the long run.

3. For improving the overall condition of African Catfish specifically, the team tailored the monitoring parameters to match what these fish need to thrive. The system kept constant watch over water quality, temperature, dissolved oxygen, and turbidity all critical factors for healthy catfish. Perhaps most valuable was the remote access feature, which let busy fish keepers check on their aquarium or pond from anywhere using their phones or computers. Whether at work, on vacation, or just in another room, users received notifications through the website whenever something needed attention. This constant connection between keeper and environment ensured the catfish remained in stable conditions even when nobody was physically present to check on them.

5.3 Recommendation

Based on the findings and conclusion of the study, these are the following recommendations that the proponent:

1. Increase the number of parameters detected by the system to enhance its monitoring capabilities beyond the current limit of five.
2. Expand the detection capabilities to include a broader range of possible causes of catfish mortality, which can significantly improve its diagnostic accuracy.

3. Implementing proper automatic water regulation features to adjust environmental conditions in real-time, reducing the need for human intervention and further safeguarding catfish health.